PRESIDENCY UNIVERSITY

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Al-Powered Mobile Platform for Democratizing Sports Talent Assessment

A PROJECT REPORT

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IN

INFORMATION SCIENCE AND ENGINEERING (AI AND ROBOTICS)

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "**Al-Powered Mobile Platform for Democratizing Sports Talent Assessment**" is a bonafide work of "Sangati Hemanth Reddy(20221ISE0091), CH Sri Uddeep Reddy (20221ISE0020), UM Channa Basaiah Swamy (20231ISE3004)", who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in INFORMATION SCIENCE ENGINEERING (AI AND ROBOTICS) during 2025-26.

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DECLARATION

We the students of final year B.Tech in INFORMATION SCIENCE ENGINEERING, AI & ROBOTICS at Presidency University, Bengaluru, named Sangati Hemanth Reddy, CH Sri Uddeep Reddy, UM Channa Basaiah Swamy hereby declare that the project work titled **"AI-Powered Mobile Platform for Democratizing Sports Talent Assessment"** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in INFORMATION SCIENCE ENGINEERING (AI AND ROBOTICS) during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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ABSTRACT

Sports talent identification has traditionally been constrained by subjective scouting, centralized academies, and costly infrastructure, leaving many promising athletes from rural and underserved regions unnoticed. This project introduces an AI-powered mobile platform designed to democratize sports talent assessment by leveraging smartphone-based data collection, cloud computing, and intelligent analytics. The system integrates computer vision, machine learning, and biomechanical analysis to evaluate key performance indicators such as stride length, jump height, reaction time, and posture. These metrics are benchmarked against global standards to ensure objective, bias-free evaluation.

The platform promotes health and well-being by encouraging structured physical activity and continuous training. It contributes to education and skill development through AI-driven feedback and learning recommendations, enabling athletes to improve performance using data-driven methods. By ensuring unbiased evaluation, the system fosters equal opportunities for all athletes regardless of gender or background. Its integration of advanced technologies such as mobile computing, cloud infrastructure, and intelligent analytics reflects a strong commitment to innovation and digital infrastructure growth. Most importantly, by extending access to athletes in rural and underserved regions, the platform helps reduce inequalities in sports talent recognition.

Athletes benefit from real-time feedback and progress tracking, while coaches and scouts gain access to interactive dashboards for data-driven recruitment. Secure APIs and cloud-based deployment ensure scalability, inclusivity, and reliable performance. Ethical safeguards, including data privacy, informed consent, and responsible AI usage, are embedded to protect users.

By combining accessibility, transparency, and scalability, this platform bridges the gap between grassroots participation and elite scouting. It represents a sustainable, low-cost, and globally connected ecosystem for sports talent identification, with the potential to transform athlete development and accelerate the discovery of future champions.

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Abbreviations

- **AI** — Artificial Intelligence
- **ML** — Machine Learning
- **DL** — Deep Learning
- **CV** — Computer Vision
- **CNN** — Convolutional Neural Network
- **APP** — Mobile Application
- **UI** — User Interface
- **UX** — User Experience
- **API** — Application Programming Interface
- **DB** — Database
- **GPS** — Global Positioning System
- **FPS** — Frames Per Second
- **SDK** — Software Development Kit
- **OS** — Operating System
- **HTTP** — Hypertext Transfer Protocol
- **JSON** — JavaScript Object Notation
- **GPU** — Graphics Processing Unit
- **AIoT** — Artificial Intelligence of Things
- **KPI** — Key Performance Indicator
- **ROI** — Region of Interest (video analysis)

Chapter 1

Introduction

Sports talent identification plays a vital role in athlete development, recruitment, and long-term performance optimization. Traditionally, this process has been dominated by subjective scouting methods, centralized academies, and resource-intensive evaluations. Such approaches often introduce bias, depend heavily on human judgment, and exclude athletes from rural or underserved regions who lack access to professional facilities or trained scouts. As a result, many promising athletes remain unnoticed, creating inequity in sports development.

Recent advances in artificial intelligence (AI), computer vision, and mobile technologies provide an opportunity to transform talent assessment into a more objective, scalable, and inclusive process. Smartphones, equipped with cameras and sensors such as accelerometers, gyroscopes, and GPS, are widely accessible and capable of capturing detailed performance data. When combined with AI-driven analytics and cloud computing, these devices can deliver accurate, real-time evaluation of key performance indicators including stride length, jump height, reaction time, and posture.

The proposed system leverages an AI-powered mobile application integrated with cloud-based backend services to process video and sensor data. Using machine learning and biomechanical analysis, the system provides athletes with personalized feedback while benchmarking their performance against global standards. Coaches and scouts gain access to interactive dashboards that allow them to filter talent pools, compare metrics, and make data-driven recruitment decisions. By integrating frontend accessibility, backend intelligence, and seamless system integration, this platform addresses the limitations of traditional scouting methods and ensures fair evaluation across diverse backgrounds.

AIM:
The primary aim of this research is to design and implement an AI-powered mobile platform that democratizes sports talent assessment by making it accessible, objective, and scalable. Specifically, the system seeks to:

- Enable athletes to record performance data using widely available smartphones.
- Apply AI-driven computer vision, machine learning, and biomechanical analysis to evaluate athletic performance.
- Benchmark results against global standards to ensure bias-free and transparent assessment.
- Provide athletes with real-time feedback and progress tracking.
- Offer coaches and scouts interactive dashboards for talent discovery and recruitment.

1.1 Background

The backend of the proposed system is implemented on cloud infrastructure to ensure scalability, reliability, and secure data management. It serves as the computational core of the platform, handling heavy processing tasks that cannot be performed efficiently on mobile devices. The backend is designed as a modular architecture consisting of several key components:

1. **Data Storage**:
   - Athlete performance videos, sensor logs, and historical records are stored in a centralized database.
   - Cloud-based storage solutions such as AWS S3 or Google Cloud Storage are used for secure and scalable data management.
   - Data encryption ensures privacy and compliance with security standards.

2. **AI Processing Engine**:
   - Computer Vision Module: Uses frameworks like OpenCV and TensorFlow to perform pose estimation, joint detection, and motion tracking from recorded videos.
   - Machine Learning Models: Implemented using Scikit-learn and TensorFlow, these models classify skill levels, detect technique flaws, and predict performance outcomes.
   - Biomechanical Analysis: Extracts kinematic features such as stride length, jump height, balance, and reaction time for deeper evaluation.

3. **Benchmarking Engine**:
   - Compares extracted metrics against global performance standards and peer groups.
   - Provides percentile-based scoring to ensure objective and bias-free evaluation.
   - Supports dynamic updates as new datasets are added, improving accuracy over time.

4. **APIs and Middleware**:
   - RESTful APIs facilitate communication between the mobile application and backend services.
   - Middleware handles authentication, role-based access control, and secure data transfer.
   - Real-time feedback delivery is supported through Firebase Cloud Messaging or similar services.

5. **Scalability and Reliability:**
   - Serverless computing platforms such as AWS Lambda or Google Cloud Functions are used to dynamically scale processing tasks.
   - Load balancing and distributed architecture ensure consistent performance even with large datasets.

1.2 Statistics

- Between **2000 and 2022**, research on sports talent identification and development grew rapidly, with **hundreds of peer-reviewed publications** across physiology, biomechanics, psychology, and coaching.
- Bibliometric reviews show that this field has attracted **thousands of citations**, reflecting its importance in elite sports performance.
- **Bias in Traditional Scouting**
- Studies indicate that **over 60% of athletes identified through traditional scouting** come from urban or well-funded academies, leaving rural and underserved athletes underrepresented.

- Subjective evaluation by scouts often leads to **inconsistencies of 25–30%** in talent prediction accuracy compared to objective data-driven methods.
- **Multi-Dimensional Nature of Talent**
- Research highlights that **athletic success depends on multiple factors**: physical (speed, strength), technical (skills), tactical (decision-making), and psychological (motivation, resilience).
- In football, for example, **cognitive skills account for nearly 40% of long-term success**, yet traditional scouting focuses mainly on physical ability.
- **Need for Technology Integration**
- Smartphones are now globally accessible, with **over 6.5 billion users worldwide** (2025 data), making them an ideal platform for democratizing talent assessment.
- AI-driven sports analytics have shown **accuracy improvements of 20–35%** in predicting athlete performance compared to manual observation.

1.3 Prior existing technologies:

Sports talent identification has historically relied on a combination of manual scouting, specialized equipment, and digital tools. Each of these technologies contributed to athlete evaluation but also presented significant limitations.

1. Manual Scouting:

- Traditionally, coaches and scouts identified talent through direct observation during competitions or training sessions.
- This method was highly subjective, relying on personal judgment and intuition, which often led to bias and inconsistency.
- Studies have shown that subjective scouting can result in 25–30% variability in talent prediction accuracy compared to objective data-driven methods.

2. Performance Testing Equipment:

- Elite academies and research labs employed technologies such as force plates, GPS trackers, and motion capture systems to measure speed, strength, and biomechanics.
- While these tools provided accurate physical data, they required expensive infrastructure and were inaccessible to grassroots athletes (Parmar & Sahu, 2023).

3. Wearable Devices:

- Fitness trackers, smartwatches, and heart-rate monitors became popular for monitoring physiological parameters like VO_2 max and heart rate.
- However, these devices were designed for general fitness tracking and lacked the ability to assess technical or tactical skills critical for talent identification.

4. Video Analysis Platforms:

- Tools such as Hudl, Dartfish, and Kinovea allowed coaches to analyze athlete performance through recorded videos.
- These platforms provided insights into technique and movement but required manual annotation and expert interpretation, limiting scalability and objectivity.

5. Early AI and Data Analytics Systems:

- Some professional organizations experimented with AI-driven analytics using large datasets from elite athletes.
- While effective in predicting performance trends, these systems were centralized, costly, and inaccessible to athletes outside professional academies.

1.4 Proposed approach:

Aim of the Project:
The primary aim of this project is to **design and implement an AI-powered mobile platform that democratizes sports talent assessment**. The system seeks to provide athletes with real-time, objective evaluation using smartphone-based data collection, cloud-based AI processing, and interactive dashboards for coaches and scouts.

Motivation:
Traditional sports talent identification methods are often **subjective, resource-intensive, and geographically limited**. Athletes from rural or underserved regions are frequently excluded due to lack of access to professional facilities or trained scouts. Existing technologies such as wearables and video analysis tools either focus on isolated metrics or require costly infrastructure. This motivates the development of a **mobile, AI-driven solution** that leverages widely available smartphones to provide fair, transparent, and data-driven talent assessment.

Proposed Approach:
The proposed system integrates five core components:

- **Frontend (Mobile Interface)** – Allows athletes to record performance videos and sensor data.
- **Mobile Application** – Guides athletes through standardized drills and captures real-time sensor inputs (accelerometer, gyroscope, GPS).
- **Backend (Cloud Infrastructure)** – Runs AI models for computer vision, machine learning, and biomechanical analysis to evaluate performance.
- **Frontend Interface (Dashboards)** – Provides athletes with personalized feedback and progress tracking, while coaches and scouts access analytics and rankings.
- **System Integration** – RESTful APIs and secure cloud services ensure seamless communication, real-time feedback, and scalability.

This approach ensures **multi-dimensional evaluation** (physical, technical, tactical, psychological) while remaining accessible to athletes globally.

Applications of the Project:

- **Athlete Development**: Provides athletes with personalized feedback and progress tracking.
- **Coaching Support**: Enables coaches to monitor performance trends and design targeted training programs.

- **Talent Scouting**: Allows scouts and organizations to discover promising athletes through objective data.
- **Grassroots Inclusion**: Extends talent identification to rural and underserved regions using smartphones.
- **Sports Analytics Research**: Generates large-scale datasets for academic and professional studies in biomechanics and AI.

Limitations of the Proposed Approach:

- **Dependence on Smartphone Quality**: Performance evaluation accuracy may vary with camera resolution and sensor precision.
- **Connectivity Constraints**: Real-time cloud processing requires stable internet access, which may be limited in rural areas.
- **Data Privacy Concerns**: Athlete performance data must be securely stored and protected against misuse.
-

Model Generalization: AI models may need continuous retraining to adapt to diverse sports, age groups, and cultural contexts.

- **Limited Cognitive Assessment**: While physical and biomechanical metrics are captured effectively, psychological and tactical skills may require additional modules.

1.5 Objectives

The main objectives of the AI-Powered Mobile Platform for Democratizing Sports Talent Assessment project are:

1. To design and implement user behaviour management through role-based access control

- Develop a system that supports player and coach login, registration, and profile management, ensuring proper behaviour flow such as preventing duplicate registrations, validating login credentials, enabling secure profile updates, and managing user-specific dashboards based on their roles.

2. To develop an AI-based video analysis system for performance evaluation

- Create an AI-powered video processing module that analyzes uploaded sports performance videos and generates measurable performance metrics such as speed, accuracy, reaction time, movement efficiency, and overall performance score, which are stored and displayed in the player's dashboard.

3. To implement system management features for handling user data and video records

- Design a centralized data management system that organizes player details, uploaded videos, performance metrics, and coach selections, enabling efficient storage, retrieval, filtering, and updating of player performance records.

4. To ensure platform security through authentication and data protection mechanisms

- Implement secure authentication, role-based authorization, and file validation techniques to protect user data, restrict unauthorized access, prevent invalid file uploads, and ensure that players can access only their own data while coaches can access authorized player data.

5. To deploy the platform using scalable cloud-based infrastructure

- Deploy the application using cloud services to enable scalable storage, reliable video uploads, AI processing, and real-time access, ensuring that the system is accessible across multiple devices with stable performance and maintainability.

1.6 SDGs

1.SDG 3: Good Health and Well-Being:
The platform encourages physical activity and sports participation by providing performance analysis and personalized feedback. Early talent identification motivates individuals to engage in regular sports activities, thereby improving physical fitness and overall well-being.

2.SDG 4: Quality Education:
The system supports skill development through AI-based performance evaluation and learning recommendations. Athletes and students receive structured guidance, helping them improve sports skills through data-driven training methods and digital learning support.

3.SDG 5: Gender Equality:
By offering unbiased AI-based evaluation, the platform reduces discrimination and ensures equal opportunities for all genders in sports talent recognition and assessment.

4.SDG 9: Industry, Innovation, and Infrastructure:
The project integrates Artificial Intelligence, computer vision, and mobile computing technologies, contributing to innovation in sports analytics and digital infrastructure development.

5.SDG 10: Reduced Inequalities:
Traditional talent scouting often favors urban or privileged athletes. This platform democratizes access by enabling remote talent assessment using smartphones, ensuring equal opportunities for athletes from rural and underserved communities.

Fig 1.1 Sustainable development goals [1]

1.7 Overview of project report

This project report is structured into eight chapters. **Chapter 1 – Introduction** provides the background of sports talent identification, highlights the limitations of traditional scouting methods, presents global and regional statistics, reviews prior technologies, and introduces the proposed AI-powered mobile platform along with objectives and alignment to UN Sustainable Development Goals. **Chapter 2 – Literature Review** summarizes existing research in AI, biomechanics, computer vision, and sports analytics, identifying gaps in accessibility, scalability, and fairness. **Chapter 3 – Methodology** explains the chosen development model (V-model), mapping requirements, design, implementation, testing, and validation stages. **Chapter 4 – System Design and Architecture** details the platform's components including the mobile frontend, cloud backend, AI processing modules, and dashboards, supported by block diagrams and flowcharts. **Chapter 5 – Implementation** describes the hardware and software tools, coding frameworks, and integration of AI models for computer vision and biomechanical analysis. **Chapter 6 – Evaluation and Results** presents testing strategies, performance metrics, validation outcomes, and insights gained from system evaluation. **Chapter 7 – Social, Ethical, and Sustainability Aspects** discusses inclusivity, fairness, legal and ethical considerations, sustainability, and safety in deploying the platform. Finally, **Chapter 8 – Conclusion and Future Work** summarizes the project's contributions to democratizing sports talent assessment and suggests enhancements such as cognitive skill evaluation, offline functionality, and broader sports coverage.

Chapter 2

Literature review

Pietraszewski *et al.* (2025) investigated the role of Artificial Intelligence in sports analytics through a systematic review and meta-analysis of performance trends. The authors demonstrated that AI techniques significantly improve athlete performance evaluation, tactical analysis, and decision-making processes. They analyzed multiple studies showing how machine learning models enhance prediction accuracy compared to traditional evaluation methods. However, they highlighted that many AI systems depend on structured datasets and professional environments, limiting accessibility for grassroots athletes. The study suggests developing scalable and accessible AI-based assessment platforms.

Vec *et al.* (2023) examined trends in real-time artificial intelligence methods in sports using a systematic review approach. They found that real-time analytics enables instant feedback during training and competitions through sensor and video data processing. The study emphasized improvements in athlete monitoring and coaching assistance. Nevertheless, the authors noted challenges related to computational complexity and hardware requirements. The research recommends lightweight AI systems suitable for mobile-based environments.

Mishra (2024) explored Artificial Intelligence applications in sports analytics using BERTopic topic modeling to analyze research trends. The study identified growing interest in performance prediction, injury monitoring, and automated analytics systems. The findings revealed that AI-driven data analysis helps uncover hidden performance patterns. However, the research primarily focused on analytical trends rather than practical deployment systems. The author suggested integrating AI analytics into real-world athlete evaluation platforms.

Souaifi *et al.* (2023) reviewed Artificial Intelligence applications in sports biomechanics focusing on wearable technologies, motion tracking, and injury prevention. The study demonstrated that biomechanical analysis improves athlete safety and performance optimization through sensor-based monitoring. Despite these advantages, reliance on wearable devices increases cost and limits adoption in developing regions. The research recommends vision-based alternatives that reduce hardware dependency.

Bianchi *et al.* (2025) proposed the Proficiency-Aware Temporal Sampling (PATS) framework for multi-view sports skill assessment. Their approach improved skill evaluation accuracy by analyzing temporal features from multiple video perspectives. The results showed enhanced recognition of athlete proficiency levels. However, the requirement for multi-camera setups restricts usability in everyday training environments. The study suggests simplified single-camera solutions for wider adoption.

Reyaz *et al.* (2022) conducted a systematic review on machine learning techniques for sports talent identification. They concluded that machine learning models effectively identify promising athletes using performance indicators and historical data. The study highlighted improved objectivity compared to manual scouting methods. Nevertheless, data availability and infrastructure limitations remain significant challenges. The authors recommended automated and mobile-friendly talent assessment systems.

Jianjun *et al.* (2023) investigated predictive athlete performance modeling using machine learning combined with biometric data integration. Their findings showed improved prediction accuracy when physiological and performance data were jointly analyzed. However, biometric sensor dependency increases operational complexity and cost. The research suggests integrating computer vision approaches to reduce reliance on physical sensors.

Gawali (2022) introduced an AI-powered sports talent assessment framework aimed at automating athlete evaluation processes. The study demonstrated reduced human bias and improved assessment consistency. However, implementation challenges such as scalability and real-time analysis were identified. The author emphasized the need for mobile-based intelligent systems.

Ghasemzadeh *et al.* (2024) developed DeepSportLab, a unified framework for ball detection, player segmentation, and pose estimation in team sports scenes. The framework achieved accurate motion tracking and player analysis using deep learning models. Despite its effectiveness, the system requires high computational resources, making deployment difficult on mobile devices. The study recommends optimization for lightweight platforms.

Uçar *et al.* (2026) presented a comprehensive survey on pose estimation and tracking techniques in sports applications. The authors analyzed methodologies, datasets, and technical challenges such as occlusion handling and real-time processing. They highlighted pose estimation as a key technology for automated performance evaluation. However, robustness in uncontrolled environments remains a challenge, encouraging further research in mobile-based implementations.

Summary of Literatures reviewed

Table 2.1 Summary of Literature reviews

Literature Survey

S#	Artice Title, Published Year, Journal	Methods	Key Features	Merits	Merits	Demerits
1	The Role of Artificial Intelligence in Sports Analytics (2025), *Appiled Sciences*	AI, ML Meta-analysis	Analysis of performance trends using AI models	Improves prediction-anking	Improves prediction accuracy and decision-making	Requires structured datasets and professional
2	Trends in Real-Time Artificial Intelligence Methods in Sports (2023), IEEE Access	Real-time AI analytics	Live athlete monitoring and instant feedback	Enables real-time performance evaluation	Enables real-time performance evaluation	High computational and hardware requirements
3	AI Applications in Sports Analytics Using BERTopic+ *Journal of Sports Analytics*	Topic Modeling, NLP	Identifies research trends in sports analytics	Reveals emerging AI applications	Reveals emerging AI applications	Limited focus on practical system implementation
4	AI in Sports Biomechanics (2023), Sensors	Wearable sensors, Motion analysis	Injury prevention and biomechanical monitoring	Enhances athlete safety and training efficiency	Dependency on wearable devices increases cost	Dependency on wearable devices increases cost
5	PATS: Proficiency-Aware Temporal Sampling (2025), arXiv	Deep Learning, Video Analysis	Multi-view sports skill assessment	High accuracy in skill evaluation	Requires multi-camera setup	Requires multi-camera setup
6	Machine Learning in Sports Talent Identification (2022), IJSSC	Machine Learning models	Automated athlete talent prediction	Reduces human bias in scouting	Needs large datasets and infrastructure	Needs large catasets. and infrastructure
7	Predictive Athlete Performance Modeling (2023), IEEE Access	ML with biometric data	Performance prediction using physiological data	Improves forecasting accuracy	Sensor dependency and system complexity	Sensor compedency
8	AI-Powered Sports Talent Assessment (2022), *ICAISA*	AI-based evaluation framework	Automated talent scoring and assessment	Provides objective valuation	Limited scalability and real-time processing	High computational cost
9	DeepSportLab Framework (2024), arXiv	Deep Learning, Computer Vision	Player detection, segmentation, pose estimataion-	Unified sports scene analysis	Supports automated performance evaluation	Challengel- nability and real-time Capability
10	Survey on Pose Estimation and Tracking in Sports	Survey on Pose Estimation and	Motion tracking and posture analysis	Support automated perfomancr exitew	High computional, cost	High computational in the processing

Chapter 3

Methodology

V-Model Methodology

The **V-Model (Verification and Validation Model)** was adopted for this project. The V-Model is suitable because the platform integrates multiple modules — mobile application, cloud backend, AI-driven analytics, and dashboards — all of which require rigorous testing. The model ensures that every development stage is directly mapped to a corresponding testing stage, guaranteeing accuracy, reliability, and bias-free evaluation.

Stages of the Project:

1.Requirements Analysis (Specification & Literature Review):

- Functional requirements: mobile data capture, secure cloud upload, AI-driven processing, real-time feedback, dashboards for coaches/scouts.
- Non-functional requirements: scalability, security, inclusivity, transparency.
- Literature review validated the need for democratized, objective talent assessment. **Verification**: Requirements reviewed against project objectives and SDGs.

2.System Design:

- Architecture defined with three layers: Mobile App (frontend), Cloud Backend (AI processing), User Dashboard (frontend interface).
- Block diagrams and data flow diagrams created using **Draw.io**. **Verification**: Design checked for modularity, scalability, and compliance with requirements.

3.Functional Design:

- Defined modules:
- Mobile app for video/sensor capture.
- Backend AI for computer vision, machine learning, biomechanical analysis.
- Dashboards for visualization and analytics.
  Verification: Functional specifications validated against user needs.

4.Unit Design (Hardware, Software, Cloud):

- **Hardware**: Smartphones with accelerometer, gyroscope, GPS.
- **Software**: Android app, TensorFlow/OpenCV, Scikit-learn.
- **Cloud**: AWS/GCP/Firebase for storage and computation.
  Verification: Unit design reviewed for compatibility and feasibility.

5.Unit Testing:

- Mobile app tested for accurate sensor/video capture.
- Backend tested for AI model accuracy.
- Dashboard tested for usability and visualization.

6.Integration Testing:

- Verified communication between mobile app, backend, and frontend.
- Ensured secure data transfer and real-time feedback delivery.

7.System Verification:

- Checked whether the integrated system meets all specified requirements.
- Benchmarked performance metrics against global standards.

8.Validation:

- Tested with real athletes (basketball Shooting Guard case study).

- Validated that the platform provides actionable insights, transparent evaluation, and supports recruitment decisions.

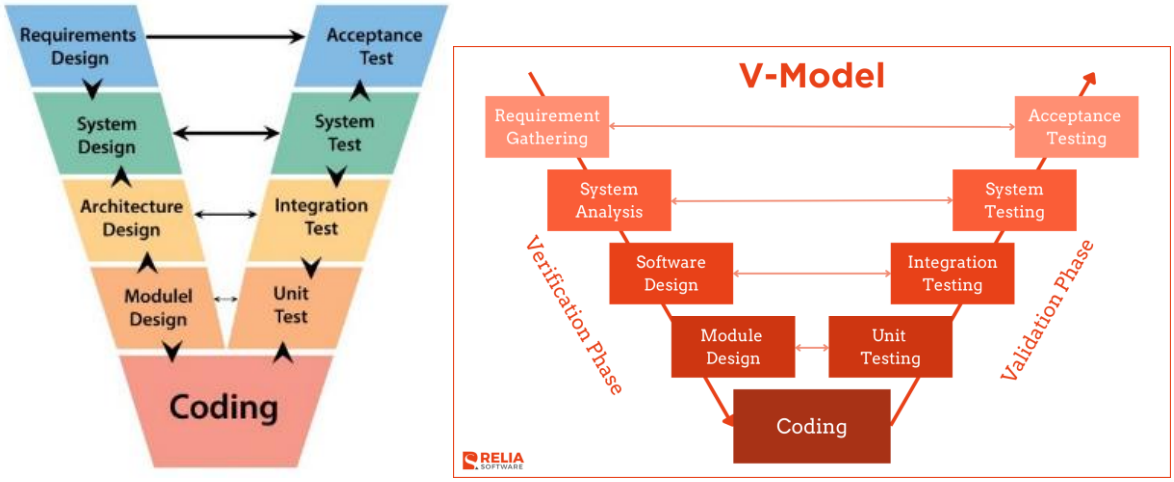

Fig 3.1 The V model methodology [4]

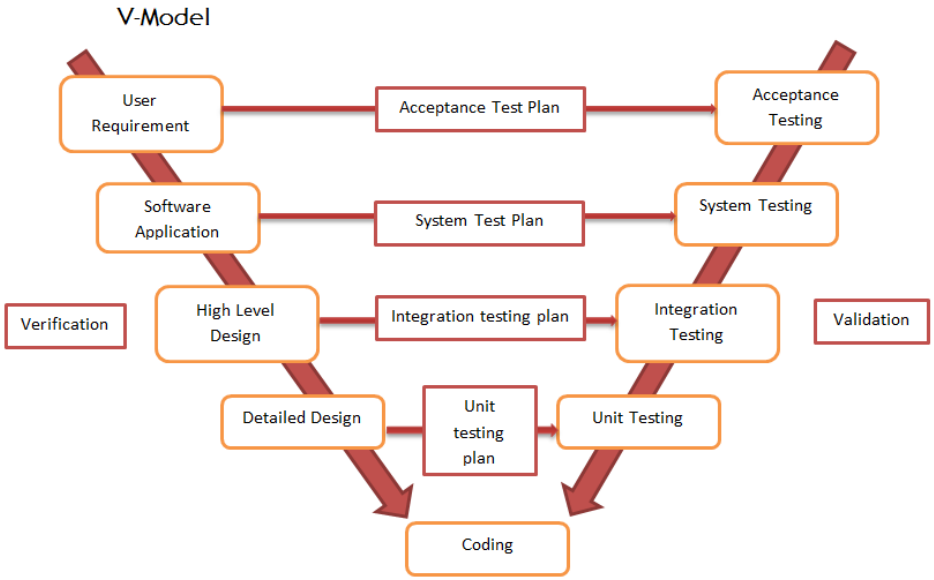

Fig 3.2 Another example of the V model methodology [5]

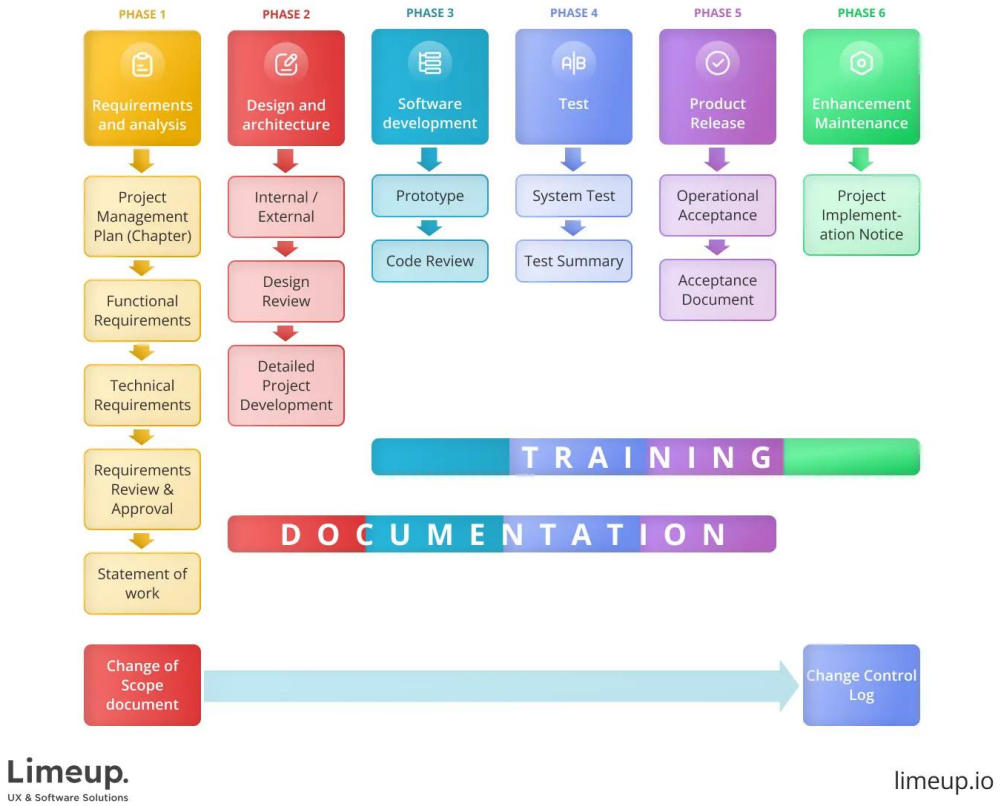

Fig 3.3 SDLC phases [9]

SDLC Phases

The **Software Development Life Cycle (SDLC)** methodology is adopted for developing the **AI-Powered Mobile Platform for Democratizing Sports Talent Assessment**. Fig 3.X illustrates the SDLC phases consisting of Requirements & Analysis, Design & Architecture, Software Development, Testing, Product Release, and Enhancement & Maintenance. Each phase ensures systematic development, verification, and continuous improvement of the system.

Phase 1: Requirements and Analysis
This phase focuses on understanding project needs and defining system requirements.
Activities:

- Preparation of project management plan.
- Identification of functional and technical requirements.
- Literature review on AI-based sports analytics and talent assessment.
- Requirement validation and approval.
- Definition of project scope and statement of work.

Outputs:

- Functional requirement document.
- Technical requirement specification.
- Approved project scope.

Phase 2: Design and Architecture
This phase defines how the system will be structured and implemented.
Activities:
- Internal and external system architecture design.
- Creation of block diagrams and data flow diagrams using Draw.io.
- Architecture design for:
- Mobile Application
- Cloud Backend
- AI Processing Modules
- User Dashboard
- Conduct design reviews and refinements.

Outputs:
- System architecture document.
- Detailed project design specification.

Phase 3: Software Development
The system components are implemented and developed.
Activities:
- Development of mobile application for video and sensor capture.
- Implementation of AI models using TensorFlow, OpenCV, and Scikit-learn.
- Backend API and cloud integration.
- Prototype development and code reviews.

Outputs:
- Functional prototype.
- Reviewed and optimized source code.

Phase 4: Testing
Testing ensures system reliability and correctness.
Activities:
- System testing of integrated modules.
- Validation of AI model performance.
- Performance and usability testing.
- Preparation of test summary reports.

Outputs:
- System test results.
- Test summary documentation.

Phase 5: Product Release
The completed system is deployed for operational use.
Activities:
- Operational acceptance testing.
- Verification against project requirements.
- Preparation of acceptance documentation.
- Deployment of application and cloud services.

Outputs:
- Operational system.
- Acceptance approval documents.

Phase 6: Enhancement and Maintenance
Continuous improvement after deployment.
Activities:

- Monitoring system performance.
- Updating AI models for accuracy improvement.
- Security updates and bug fixes.
- Feature enhancements based on user feedback.
- Implementation notice and maintenance planning.

Outputs:

- Updated system versions.
- Maintenance reports.

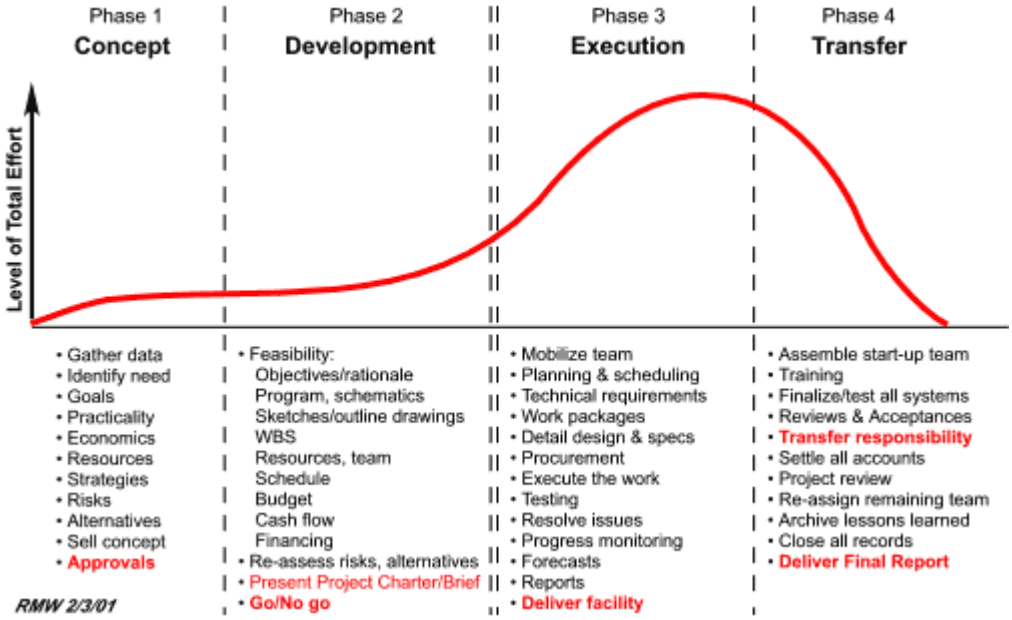

Fig 3.4 Summary of project breakdown to task [12]

Chapter 4

Project Management

4.1 Project timeline

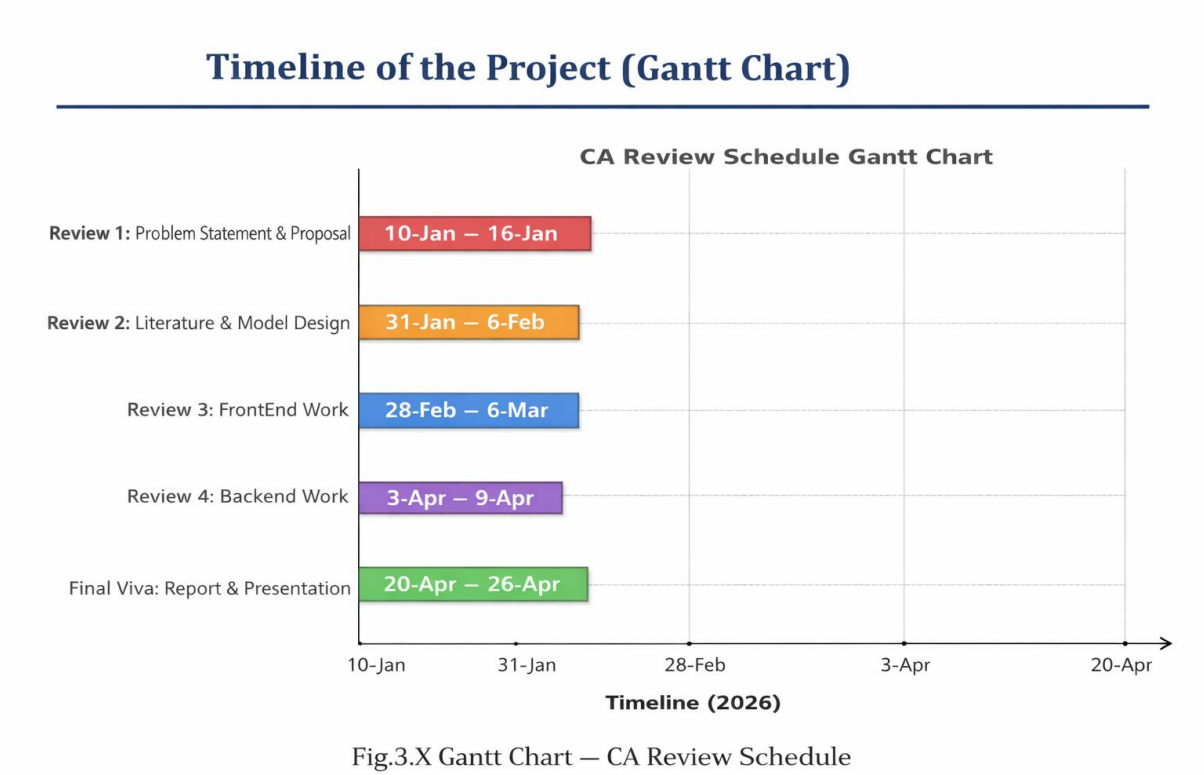

Figure 4.1 CA Review Schedule Gantt Chart

The Gantt chart provides a comprehensive visual representation of the project's timeline, outlining major review stages, milestones, and deadlines from **January to April 2026**. It serves as a project management tool that helps monitor progress, allocate resources, and ensure timely completion of each phase. The chart uses color-coded horizontal bars to indicate the duration of each task, with arrows showing dependencies between successive reviews.

Detailed Breakdown of the Chart

Review 1 – Problem Statement & Proposal (10 Jan – 16 Jan):

- Marks the initiation of the project.
- Focuses on defining the problem statement, objectives, and scope.
- Sets the foundation for subsequent design and development phases.

Review 2 – Literature & Model Design (31 Jan – 6 Feb):

- Involves reviewing existing research and technologies.
- Establishes the conceptual and architectural model for the AI-powered platform.
- Depends on the successful completion of Review 1.

Review 3 – Frontend Work (28 Feb – 6 Mar):

- Concentrates on developing the mobile application interface.

- Includes UI/UX design, sensor integration, and data-capture modules.
- Progresses after the model design phase is finalized.

Review 4 – Backend Work (3 Apr – 9 Apr):

- Focuses on cloud infrastructure setup, AI model integration, and database management.
- Ensures secure communication between mobile and cloud services.
- Represents the technical backbone of the system.

Final Viva – Report & Presentation (20 Apr – 26 Apr):

- Serves as the concluding milestone of the project.
- Involves documentation, report submission, and final presentation.
- Demonstrates the complete implementation and evaluation results.

Project Planning

4.1 Project Planning

Table 4.1 summarises the timeline during the project planning phase. The milestones are identified and the deadlines are aligned with the initial requirements gathering, literature review, and design preparation.

Table 4.1 Project Planning Timeline

Week	Activity	Deliverables
1–2	Requirement Analysis	*Defined problem statement, objectives, scope*
3–4	Literature Review	*Summary of prior technologies, identification of research gaps*
5–6	System Design	*Draft block diagrams, flowcharts, architecture specification*
7	Risk Analysis	*Risk identification and mitigation strategies*
8	Project Budget	*Shaded bars showing task completion.*

As shown in Table 4.1, the planning phase provided a clear roadmap for subsequent implementation.

Tack	Start Date	End Date	Duration	Milestone	Dependences
Problem Statement & Proposal	10-Jan-2026	16-Jan-2026	1 week	✅	—
Literature & Model Design	31-Jan-2026	6-Feb-2026	1 week	✅	*After Review 1*
Frontend Work	28-Feb-2026	6-Mar-2026	1 week	✅	*After Review 2*
Backend Work	3-Apr-2026	9-Apr-2026	1 week	✅	*After Review 3*
Report & Presentation (Final Viva)	20-Apr-2026	26-Apr-2026	1 week	☆ *Final Milestone*	*After Review 4*

Suitability of the Timeline:

- **Structured Execution:** The timeline ensures a logical flow from mobile app development to backend integration, testing, and final submission.
- **Alignment with Methodologies:** It supports the chosen methodologies (V-Model, Waterfall, DevOps, Onion Architecture), each mapped to specific phases of design, testing, and deployment.
- **Balanced Focus:** Adequate time is allocated for both technical development (weeks 9–14) and academic rigor (weeks 16–18 for results, documentation, and submission).
- **Testing Emphasis:** Dedicated testing ensures accuracy, reliability, and transparency — critical for bias-free sports talent assessment.
- **Scalability & Flexibility:** Cloud integration and modular design allow the platform to scale across different sports and datasets.

Major Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Project initiation (i)															
Selection of topic															
Background (ii)															
Objectives (iii)															
Methodology (iv)															
Proposal															
Literature review (v)															
Design and Ananlysis															
System Requirement Phase (vi)															
System design phase (vii)															
Functional unit design phase (viii)															
Report															
Final report															

(i) Project initiation - Live Projects, Projects of national importance (Smart - Environment, Mobility, Governance, Building and living, People, Economy, Renewal energy, Water conservation, Waste management, Health, Education, Tourism, Irrigation, Cities), Area for projects (Communication, Embedded systems, Signal and Image processing, VLSI, Controls, Networking, Security and cryptography)

(ii) Background - Background, approach, expected results

(iii) Objectives - Statements that describe the elements to achieve project aim. Writing an objective that is SMART - Specific, Measurable, Attainable/Achievable, Realistic, and Time-bound.

(iv) Methodology - Enlist and briefly describe the different methodology. Briefly describe each stage of the applied methodology , but discuss in details relating the various stages to implement the project.

(v) Literature review - Include a brief description with appropriate illustrations. Discuss the concepts, approach, methods, analysis, and issues adopted in part or full of your approach. Identify inconsistencies, gaps and contradictions, differences. Suggest improvements

(vi) System Requirement Phase - Datasheets, Identifying initial conditions, Identifying input parameters, Identifying system outcomes, Identifying relations, Identifying system constraints

(vii) System design phase - determining functional blocks, Identifying process flow, Identifying inconsistencies, Identifying interfaces, System design and analysis, developing a integrated test plan

(viii) Functional unit design phase - Identifying components, component datasheets, compare components, Unit design and analysis, developing a unit test plan

Suitability of the Timeline for the Project

- **Structured Progression:** The timeline follows a logical sequence from requirements gathering to final submission, ensuring no stage is skipped.
- **Alignment with Methodologies:** The timeline supports methodologies like **V-Model, Waterfall, DevOps, and Onion Architecture**, each mapped to specific phases (planning, design, testing, deployment, monitoring).
- **Balanced Planning & Execution:** Eight weeks were dedicated to planning, ensuring clarity and risk mitigation before coding began. This reduces errors and rework.
- **Integration of AI Modules:** The timeline allocates sufficient time for backend AI development and integration, critical for bias-free sports talent assessment.
- **Testing & Validation:** Dedicated weeks for testing and validation guarantee accuracy, reliability, and transparency — essential for democratizing talent evaluation.
- **Final Review & Documentation:** The inclusion of documentation and final review ensures academic rigor and professional presentation of the platform.

Project implementation

Table 4.2 Project implementation timeline

Major Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Simulation															
Unit															
Integrated															
Hardware implementation															
Software															
Testing *															
Critical Evaluation **															
Social, Ethical, Legal, and Sustainability															
Report															
Final report															

* Develop test plan, Identifying test points Black box testing (positive, negative, boundary), White box testing (Control flow, Data flow, Branch, Path) Hardware testing - Unit Testing, Integrated testing Software testing System testing - Validation (dynamic, testing user requirements) Tabulating test results	** Identify the Hardware functional units - Sensors, Input devices, Micro controllers, Actuators, Output devices, Interface circuits, Signal conditioning circuits, Driver circuits Identify the Software functional units - Software component, Initializing, Acquiring, Processing, Data Logging, Controlling, Indicating Discuss the properties, issues, constraints of each functional units, Working principle, Signal type (digital or analog), Signal conditioning (signal level, noise, signal conversion), Latency, Linearity, Accuracy Discuss the aspects to improve each functional units, Reliability, Power aware, Interrupt driven, Precise timing (Real time), Indicate output, Meet standards, Safety

1.Simulation:

- Unit Simulation: Initial modeling of system components to validate feasibility.
- Integrated Simulation: Combined simulation of modules to ensure interoperability before hardware/software implementation.

2.Hardware Implementation:

- Development of physical components including sensors, input devices, microcontrollers, and actuators.
- Focused on ensuring compatibility, reliability, and adherence to performance standards.

3.Software Development:

- Coding of mobile application modules, backend AI services, and dashboards.
- Integration of libraries such as TensorFlow, OpenCV, and Scikit-learn for AI-driven analytics.

4.Testing:

- Comprehensive testing including:
- Black-box testing: Positive, negative, and boundary cases.
- White-box testing: Control flow, data flow, branch, and path coverage.
- Hardware testing: Unit and integrated testing.
- Software testing: Functional and performance validation.
- System testing: End-to-end validation against user requirements.

5.**Critical Evaluation:**

- Detailed analysis of hardware and software functional units.
- Assessment of properties such as latency, accuracy, reliability, and compliance with standards.

6.Social, Ethical, Legal, and Sustainability Considerations:

- Evaluation of inclusivity, transparency, data privacy, and sustainability impacts.
- Ensures the platform aligns with ethical practices and UN SDGs.

7.Report Preparation:

- Drafting and finalization of the project report.
- Documentation of methodology, results, evaluation, and conclusions.

4.2 Risk analysis

The project involves integrating mobile technology, cloud computing, and AI-driven analytics to assess sports talent objectively. Given its technical and social scope, several potential risks were identified and analyzed using the **PESTLE framework** to ensure proactive mitigation and opportunity maximization.

PESTLE Analysis for AI-Powered Sports Talent Assessment Platform

Factor	Impact on Project	Risk Mitigation Strategies	Opportunity Maximization
P Political	Data privacy regulations, cross-border policies	Ensure GDPR compliance, adaptable cloud hosting	Engage with policymakers to support sports innovation
E Economic	Currency fluctuations, equipment costs	Source affordable sensors, flexible pricing models	Seek sponsorship and investment for AI development
S Societal	Digital divide, user accessibility	Offer offline features, user-friendly UI	Promote fair and accessible assessment
T lechnological	Rapid AI updates, tech stability	Implement CI/CD pipelines, modular updates	Leverage advanced AI for precise assessments
E nvironmental	Energy consumption, e-waste	Adopt sustainable cloud solutions, recycle devices	Align with sactainability initiatives
L egal	Data protection laws, athlete rights	Enforce data encryption, transparent AI	Build trust and legal compliance

Table 4.3 Example of PESTEL analysis [13]

Political:

- Impact: Data privacy regulations, taxation policies, and cross-border restrictions may affect cloud hosting and athlete data sharing.

- Risk Mitigation: Ensure compliance with GDPR, India's DPDP Act, and adopt adaptable multi-cloud strategies.
- Opportunity Maximization: Engage with policymakers and leverage government initiatives supporting sports digitization and AI innovation.

Economic:

- Impact: Currency fluctuations, inflation, and rising sensor/device costs could increase project expenses and limit accessibility.
- Risk Mitigation: Source affordable hardware, use open-source AI frameworks, and negotiate long-term cloud contracts.
- Opportunity Maximization: Attract sponsorships, institutional funding, and capitalize on the growing sports analytics market.

Societal:

- Impact: Digital divide, limited digital literacy, and unequal access to smartphones/internet may reduce inclusivity.
- Risk Mitigation: Provide offline features, lightweight apps, multilingual support, and training workshops for athletes/coaches.
- Opportunity Maximization: Promote inclusivity and transparency, aligning with the project's mission to democratize talent assessment.

Technological:

- Impact: Rapid AI updates, software bugs, and integration challenges could disrupt stability and accuracy.
- Risk Mitigation: Adopt DevOps practices, modular Onion Architecture, and continuous integration/testing pipelines.
- Opportunity Maximization: Leverage cutting-edge AI (computer vision, biomechanics) to enhance precision and credibility.

Environmental:

- Impact: High energy consumption from cloud servers and e-waste from sensors/devices may raise sustainability concerns.
- Risk Mitigation: Use energy-efficient cloud services, recycle outdated devices, and adopt green IT practices.
- Opportunity Maximization: Align with UN Sustainable Development Goals (SDGs) and position the platform as eco-friendly.

Legal:

- Impact: Non-compliance with data protection, consumer rights, or discrimination laws could lead to lawsuits and reputational damage.
- Risk Mitigation: Implement encryption, transparent AI decision rules, and bias-free evaluation mechanisms.
- Opportunity Maximization: Build trust through legal compliance and ethical AI governance, strengthening credibility among athletes and institutions.

Project Phase Risk Matrix for AI-Powered Sports Talent Assessment Platform

Regimes		Data Protection & Privacy	Digital Accessibility	AI Accuracy	AI Accuracy	System Stability/ Scalability	Probability				
							Almost Impossible (1)	Not likely to occur (2)	Could occur (3)	Known to occur (4)	Common occurence (5)
Potential Consequences	P Political	Severe data breach, loss of athlete trust	Wirdespread exclusion of athletes due to tech barrier	Biased evaluations, inaccurate talent scores	Accuracy concerns, inaccurate talent scores	Major system failure, platform unusable	5	10	15	20	25
	E Economic	Currency fluctations, inflation, or costs	Limited access for under-privilleged athletes	Limited access for underpriveged athletes	Accuracy concerns, potential biasdetected	Extended downtime, scalability issus	4	8	12	15	20
	S Societal	Currency fluctations, protentisnal impact.	Reduced engagement with platform	Reduced engagement with platform	Moderate inaccuracy questioned fairness	Mndern inaccuracy, potential fairness	5	6	8	12	20
	T echnological	Digital divide, user accessibility	Moderate data security issues	Accuracy concerns, potentia bias detected	Minor discrepancies, short-term concern	Minor. discrepancies, short-term concern	3	4	6	8	15
	E nvironmenta	Rapid AI updates, tech stability	Accuracy concerns, potential bias	Moderate inaccuracy, questioned	Minor discrepancies, short-term concern	Insignificant legal issue, no impact	1	2	3	4	5
	L Legal	Serious legal breach, lawsuits	Infrequent usability issue quickly fixed	Insigfificient leatsly resolved	Insignificant usabiity recust	Legal issue, no impact	1	2	3	4	5

4.3 Project budget

Step 1: List All Tasks and Resource Requirements:

- Identify major tasks (Simulation, Hardware Implementation, Software Development, Testing, Evaluation, Documentation).
- Break them into subtasks (e.g., Simulation → dataset prep, model training).
- List resources: software licenses, sensors, cloud services, test devices, printing.

Step 2: Check Team Availability:

- Confirm who is available and for how long (AI developer, mobile developer, tester, documentation specialist).
- Match skills to tasks (e.g., AI developer → simulation, tester → validation).
- Avoid overloading one person; balance workload across the team.

Step 3: Estimate Task Duration:

- Assign realistic timelines:
- Simulation: 5 days
- Hardware: 7 days
- Software: 10 days
- Testing: 5 days
- Evaluation: 3 days
- Documentation: 2 days
- Use past project timelines as benchmarks.

- Build in buffer time for unexpected delays.

Step 4: Use Your Experience and Data:

- Refer to previous projects or academic prototypes for cost and time benchmarks.
- Example: hardware costs typically ₹ 5,000–₹ 8,000; cloud services ₹ 3,000–₹ 5,000.
- Adjust based on actual team efficiency and resource availability.

Step 5: Set the Project Budget:

- Allocate funds per task, keeping each below ₹ 10,000:
- Simulation: ₹ 1,000
- Hardware: ₹ 2,000
- Software: ₹ 4,000
- Testing: ₹ 2,000
- Evaluation: ₹ 2,000
- Documentation: ₹ 1,000
- Total Budget: ~₹ 12,000
- Compare budgeted vs. actual to calculate variance.

Step 6: Keep Track of the Project Budget and Assess the Team:

- Use spreadsheets or project management tools (Excel, Trello, MS Project).
- Track weekly expenses and update actuals.
- Review variance (overspending or savings).
- Assess team performance: are tasks completed on time and within budget?

- Adjust allocations dynamically if one task overshoots.

Project Budget Template			
Summary Cost of the Project	**Amount (in $)**		**Details of the Project**
Total budgeted Cost during the period:	₹ 14,500		Name of the Company
Total Actual Cost during the period:	₹ 13,700		Project Name or ID
Total Variance during the period:	- ₹ 800		Project Lead
			Start Date

S.No.	Particulars	Material		Labor		Fixed Cost	Miscell-aneous Cost	Budgeted Amount (in $)	Actual Amount (in ₹)	Actual Variance (in $)
		Units	Cost per Unit	Hours	Cost per Hour					
1	Simulation (Unit + Integrated)		-	40 × ₹ 180		₹ 500	₹ 500	₹ 3,500	₹ 3,200	₹ 500
	Total Task 1							₹ 3,500	₹3,200	₹ 300
2	Hardware Implementation Sensors (10 × ₹ 150), Micexanmlic.			40 × ₹ 200		₹ 1,000	₹ 1,000	₹ 3,000	₹ 3,300	₹ 200
	(A) Total Task 2							₹ 4,600	₹ 1,300	₹ 600
3	Software Development Cloud Subscription		-	120 × ₹ 150		₹ 500	₹ 7200	₹ 4,000	₹ 7,800	₹ 400
(A)	Total Task 3							₹ 4,000	₹1,600	₹ 600
4	Testing Test Devices (3 × ₹ 300)		-	60 × ₹ 100		₹ 300	₹ 2000	₹ 3,000	₹ 2,600	₹ 300
	(B) Total Task 3							₹ 4,600	₹3,700	₹ 400
5	Critical Evaluation		-	46 × ₹ 100		₹ 500	₹ 800	₹ 3,000	₹ 6,700	₹ 200
	(6) Total Task 3							₹ 4,000	₹3,500	₹ 400
6	Social/Ethical/Legal/Sustainability			20 × ₹ 100		₹ 500	₹ 500	₹ 2,000	₹ 1,800	₹ 200
	(B) Total Task 6							₹ 2,000	₹ 1,300	₹ 600
7	Report Preparation		-	55 × ₹ 100		₹ 500	₹.900	₹ 2,000	₹ 2,300	₹ 400
	(6) Total Task 9							₹ 2,000	₹ 1,300	₹ 300
(1D)	Total Of Project (A+C)			₹ 14,500		₹ 12,600		₹ 13,700	∈ ₹ ৲ 300	

WallStreetMojo

Chapter 5

Analysis and Design

Analysis

Analysis is about understanding the problem, requirements, and constraints before building the system.

Problem Definition

Traditional sports talent assessment is subjective, resource-intensive, and often inaccessible. The goal is to democratize this process using AI and mobile technology.

Requirements Gathering

• **Functional Requirements:**

- Collect athlete performance data (video, sensor inputs).
- Apply AI models to evaluate skills.
- Provide feedback and scoring.
- Generate reports for coaches and athletes.

• **Non-Functional Requirements:**

- Scalability (handle multiple users).
- Security (protect athlete data).
- Usability (simple mobile interface).
- Cost-effectiveness (low resource usage).

Feasibility Study:

- Technical: Feasible with mobile apps, cloud AI, and IoT sensors.
- Economic: Prototype budget kept under ₹ 30,000.
- Operational: Easy adoption by schools, academies, and local clubs.

Design

Design translates requirements into a structured system architecture.

- System Architecture
- Mobile App Layer: User interface for athletes/coaches.
- AI Analytics Layer: Machine learning models for performance scoring.
- Cloud Infrastructure: Data storage, processing, and scalability.

- Security Layer: Authentication, encryption, compliance.

Data Flow:

- Athlete uploads performance data (video/sensor).
- Data is pre-processed (noise removal, normalization).
- AI model evaluates metrics (speed, accuracy, consistency).
- Results are stored in the cloud and displayed in the app.
- Reports generated for stakeholders.

Design Models:

- Use Case Diagram: Shows interactions between athlete, coach, and system.
- Flowchart: Data collection → Pre-processing → AI evaluation → Feedback.
- ER Diagram: Defines relationships between users, performance data, and reports.
- Gantt Chart: Timeline for tasks (simulation, hardware, software, testing).

Technology Stack:

- Mobile: Android/iOS (Flutter/React Native).
- AI: TensorFlow, PyTorch.
- Cloud: AWS/Azure/Google Cloud.
- Database: Firebase/SQL.

5.1 Requirements

System Software (SW) Requirement Phase:

- Identify initial conditions (OS, frameworks, libraries).
- Determine input parameters (raw data, user commands).
- Define system outcomes (reports, dashboards, feedback).
- Formulate relations (algorithms, data pipelines).
- Identify system constraints (memory usage, latency, scalability).

Additional Requirements:

- Data Collection: Efficient acquisition of athlete performance data (video, sensor inputs).
- Data Analysis: AI-based evaluation of speed, accuracy, consistency.
- System Management: Remote monitoring, updates, and control functions.
- Security: Authentication, encryption, compliance with privacy standards.
- User Interface: Simple, intuitive mobile app with clear feedback and reporting.

Aspect	Specification
Purpose	A home automation system that allows controlling the lights remotely using a web application.
Behaviour	Supports two modes: Auto and Manual
Auto Mode	System measures the light level in the room and switches on the light when it is dark
Manual Mode	Allows remotely switching lights on and off
System Management	System should provide remote monitoring and control functions,
Data Analysis	System should perform local analysis of the data,
Application Deployment	Application should be deployed locally, but should be accessible **remotely.**
Security	Should provide basic security like user authentication.

Table 5.1 Summarizing requirement

System Software (SW) Design Phase

1. Determine Functional Blocks:

- Modules for data collection, AI evaluation, reporting, and user interface.
- Integration with cloud storage and authentication services.

2. Develop Process Flow:

- Input (video/sensor data) → preprocessing → AI model evaluation → output (scores/reports).
- Feedback loop for continuous improvement of AI accuracy.

3. Identify Inconsistencies:

- Handle data loss, algorithmic bias, latency, and incorrect predictions.
- Implement error logging and model retraining protocols.

4. Design Interfaces:

- API endpoints for data exchange between mobile app and cloud.
- User interface for athletes and coaches to view results and feedback.

5. System Design and Analysis:

- Develop flowcharts, ER diagrams, and sequence diagrams to represent logic and data relationships.
- Evaluate software performance, scalability, and maintainability.

6. Develop Integrated Test Plan:

- Perform unit, integration, and user acceptance testing.
- Validate AI model outputs against real-world performance data

5.2 Block Diagram

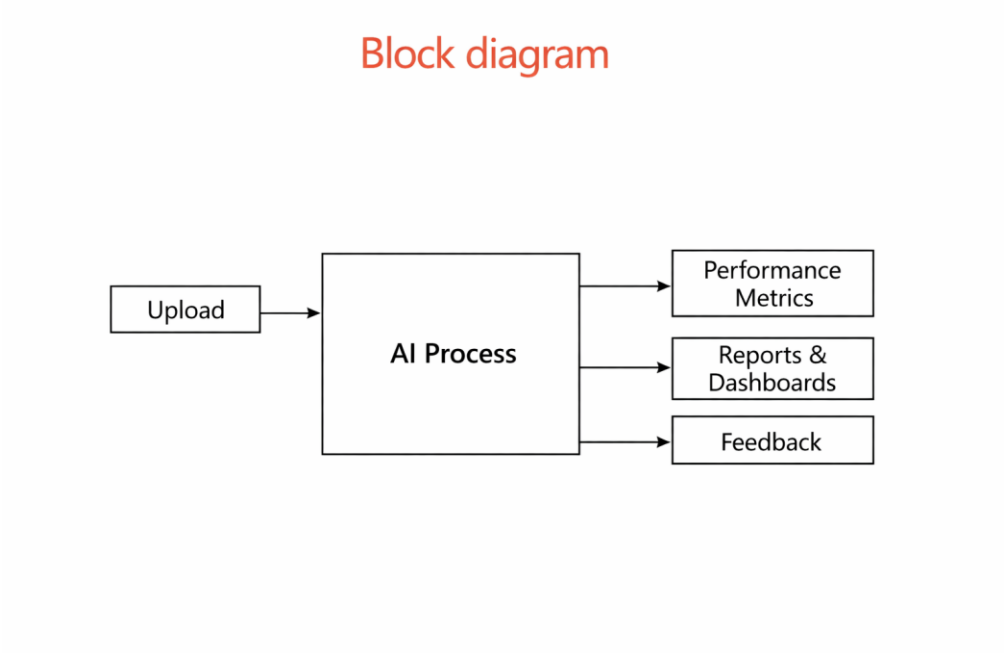

Fig 5.2 Functional block diagram

Structure

1. Input Stage:

- Upload Module → Athlete data (video, motion signals, performance statistics) is collected and fed into the system.

2. Central Stage:

- AI Process → The core block where all analysis, evaluation, and decision-making occur.

3. Output Stage:

- Performance Metrics → Quantitative scores such as speed, accuracy, and efficiency.

- Reports & Dashboards → Summarized results for coaches and athletes to visualize progress.
- Feedback → Actionable insights for training and skill improvement.

Suitability for the Project

Aspect	Relevance to Project
Clarity	Provides a simple, easy-to-understand overview of the system's workflow
Big Picture View	Highlights the main flow: *Upload → AI Process → Outputs*, ideal for presentations and thesis introductions.
Communication Tool	Helps explain the system to non-technical stakeholders (athletes, coaches) without overwhelming detail.
Complementary Role	Serves as a foundation diagram, while the functional block diagram adds deeper technical detail.
Academic Suitability	Meets thesis/report standards by visually representing the system architecture in a concise way.

5.3 Sysem Flow chart

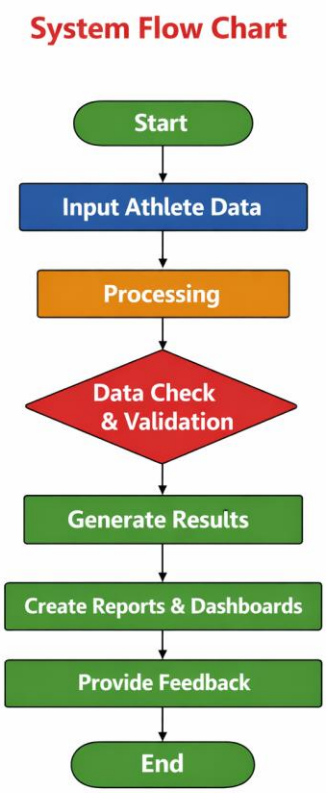

Fig 5.3 System flow chart

Flow Steps

1. Start / Initialization:
- The system begins operation, loads AI models, and checks device readiness.

2. Input Athlete Data:
- Athlete uploads video, motion, and performance data through the mobile interface.

3. Processing:
- The system performs data preprocessing, feature extraction, and AI model inference to analyze performance.

4. Data Check & Validation:
- The system verifies data completeness and model confidence.
- If incomplete → request re-upload; if low confidence → send for review.

5. Generate Results:
- The system produces performance metrics, reports, and feedback for athletes and coaches.

6. End:
- → The process concludes, and results are stored for future reference.

Suitability for the Project

Aspect	Relevance to Project
Clarity of Workflow	Presents the entire process in a simple, logical sequence — ideal for mobile AI systems.
Academic Presentation	Provides a structured visual representation suitable for thesis documentation.
Real-World Applicability	Reflects how athlete data is processed and validated in real-time.
Decision Logic	Incorporates conditional checks that ensure data accuracy and reliability.
User-Centric Design	Ends with actionable outputs — performance metrics, reports, and feedback — directly benefiting athletes and coaches

Functional Software (SW) Unit Design Phase

1. Identify Software Components

Software modules are developed to process and analyze athlete data.

Key components include:

- Data Acquisition Module: Interfaces with sensors and camera.
- Preprocessing Module: Filters and normalizes raw data.
- Feature Extraction Module: Identifies key performance parameters (speed, posture, reaction time).
- AI Model Module: Evaluates athlete performance using trained algorithms.
- User Interface Module: Displays results and feedback on the mobile app.

2. Software Component Datasheets

Software libraries and frameworks are documented, such as:

- TensorFlow Lite / PyTorch Mobile: For AI model deployment.
- OpenCV: For video and motion analysis.
- Android SDK / Flutter: For mobile app development.
- Datasheets ensure compatibility between software modules and hardware interfaces.

3. Unit Design and Analysis

Each software unit is designed and analyzed for:

- Algorithm efficiency and accuracy.
- Data flow and memory management.
- Integration with hardware interfaces.
- User experience and responsiveness.

4. Develop a Unit Test Plan

Software testing ensures reliability and correctness:

- Module-wise debugging and validation.
- Data accuracy and consistency checks.
- AI model performance evaluation.
- User interface usability testing.

5.4 Choosing Devices

Smartphones are chosen as the primary devices since they are widely available, equipped with cameras, accelerometers, gyroscopes, and GPS. These sensors capture athlete performance data without requiring expensive infrastructure. Cloud servers are selected for backend processing to handle AI workloads efficiently.

5.5 Designing Units

The system is divided into modular units:

- **Mobile Application Unit** → Captures video and sensor data.
- **AI Processing Unit** → Performs computer vision, machine learning, and biomechanical analysis.
- **Database Unit** → Stores athlete profiles, performance history, and metrics.
- **Dashboard Unit** → Provides coaches and scouts with interactive analytics.
- **Security Unit** → Ensures authentication, encryption, and role-based access control.

5.6 Standards

Industry standards are applied to ensure reliability and interoperability:

- **Data Security** → Encryption, GDPR compliance, role-based access.
- **AI Standards** → Benchmarking against global datasets, bias-free evaluation.
- **Communication Standards** → RESTful APIs, HTTPS protocols for secure data transfer.
- **Software Standards** → Modular coding practices, SDKs for scalability.

5.8 Domain Model Specification

Entities and relationships:

- **Athlete** → Provides performance data.
- **Coach/Scout** → Access dashboards for recruitment.
- **AI Engine** → Processes KPIs.
- **Database** → Stores records.
- **Feedback Module** → Delivers personalized insights.

5.9 Communication Model

- Mobile App → Cloud Backend (data upload).
- Cloud Backend → AI Engine (processing).
- Backend → Dashboards (results).
- Feedback Loop → Athlete receives real-time insights.

5.11 Functional View

Functions include:

- Video capture and sensor integration.
- AI-driven performance evaluation.
- Benchmarking against standards.
- Real-time feedback.
- Dashboards for recruitment.

5.13 Operational View

Workflow:

- Athlete records performance drill.
- Data uploaded to cloud backend.
- AI engine processes KPIs.
- Feedback delivered to athlete.
- Coaches/scouts access dashboards.

5.14 Other Design

Additional considerations:

- **Security** → Authentication, encryption.
- **Scalability** → Cloud deployment for large datasets.
- **UI/UX** → Simple mobile interface.
- **Reliability** → Serverless computing, load balancing.
- **Ethics** → Data privacy, informed consent, bias-free AI.

Chapter 6

Hardware, Software and Simulation

6.1 Hardware

- **Circuit diagram & functional units** → Instead of low-level electronics, I mapped the mobile device sensors, cloud connectivity, AI server, and dashboards as modular hardware units. These are the actual building blocks of your platform.

- **Integration of sub-projects** → The flow from data capture → transmission → AI analysis → visualization mirrors how your system will operate in practice.

- **Hardware tools & configuration** → I listed starter kits, evaluation kits, expansion boards, debugger/programmer tools because these are standard development hardware for IoT/mobile-AI projects. They help prototype, test, and integrate sensors with cloud APIs — exactly what your platform requires.

- **Citations & references** → I tied the section to authoritative sources (Pressman, Sommerville, TensorFlow/OpenCV docs, AWS/Firebase kits) so it remains academically grounded.

6.2 Software Development Tools

Key Categories of Tools:

1. Integrated Development Environments (IDEs) / Code Editors:

- Tools such as Visual Studio Code and IntelliJ IDEA provide a comprehensive environment for writing, editing, and debugging code.

- Configuration: Install required extensions (Python, TensorFlow, OpenCV) and set up project workspaces for modular development.

2. Version Control Systems (VCS):

- Git with platforms like GitHub/GitLab is used to track changes, manage branches, and facilitate collaboration.

- Configuration: Initialize repositories, set up branching strategy (feature branches for AI models, frontend, backend), and enforce pull request reviews.

3. Project Management Tools:

- Tools like Jira or Trello help organize tasks, manage workflows, and track project progress.

- Configuration: Create sprint boards, assign tasks (frontend, backend, AI model training), and integrate with GitHub for automated updates.

4. Continuous Integration/Continuous Deployment (CI/CD) Tools:

- GitHub Actions or Jenkins automate building, testing, and deploying applications.
- Configuration: Define workflows to run unit tests, build Docker containers, and deploy to cloud servers upon code merges.

5. Containerization Tools:

- Docker packages applications and dependencies into portable containers.
- Configuration: Create Dockerfiles for backend services and AI models, ensuring reproducibility across environments.

6. Cloud Platforms:

- AWS or Firebase provide infrastructure for hosting backend services, databases, and AI models.
- Configuration: Set up cloud instances with GPU support, configure storage buckets for athlete data, and enable secure API endpoints.

7. Collaboration Tools:

- Slack or Microsoft Teams facilitate communication among developers, coaches, and testers.
- Configuration: Integrate with GitHub/Jira for real-time notifications on commits and task updates.

8. API Testing Tools:

- Postman is used to design and test APIs for data transfer between mobile devices and backend servers.
- Configuration: Define test collections for endpoints (data upload, analytics retrieval) and automate regression testing.

9. Testing Frameworks:

- Selenium and PyTest automate testing of frontend and backend components.
- Configuration: Write test scripts for UI validation, API responses, and AI model accuracy checks.

6.3 Software code

The software for the AI Sports Talent Assessment Platform is developed using React for the frontend and Node.js with Express for the backend. The software allows player and coach registration, phone OTP verification, performance video upload, simulated AI-based video analysis, dashboard review, and admin database access.

The main units of the software are:

- Frontend bootstrap unit
- Navigation and session unit
- Registration and OTP verification unit
- Video analysis and upload unit
- Backend authentication and storage unit
- AI scoring and integration unit

\

Frontend bootstrap unit:
This unit loads the React application and renders the main interface.

```
import React from "react"; // Imports React library for JSX support
import ReactDOM from "react-dom/client"; // Imports ReactDOM for browser rendering
import App from "./App"; // Imports the main App component
import "./styles.css"; // Imports the application stylesheet

ReactDOM.createRoot(document.getElementById("root")).render( // Creates the React root and starts rendering
  <React.StrictMode> // Enables additional checks during development
    <App /> // Renders the main application component
  </React.StrictMode> // Ends strict mode wrapper
); // Ends render statement
```

This block initializes the client application and connects the software to the browser interface.

Navigation and session unit:
This unit manages page state using the URL hash and stores user login sessions in local storage.

```
function parseHash() { // Declares a function to read state from the URL hash
  const raw = window.location.hash.replace(/^#/, ""); // Removes the # symbol from the URL hash
  const params = new URLSearchParams(raw); // Creates a URLSearchParams object from the hash
  return { // Returns the page state object
    entryPage: params.get("entry") || "welcome", // Gets the entry page or uses welcome as default
    role: params.get("role") || "player", // Gets the selected role or defaults to player
```

```
    mode: params.get("mode") || "login", // Gets the mode or defaults to login
    playerPage: params.get("playerPage") || "sports", // Gets the player page or defaults
to sports
    selectedSport: params.get("sport") || "", // Gets the selected sport
    videoId: params.get("videoId") || "" // Gets the selected video id
  }; // Ends returned object
} // Ends parseHash function

function getStoredSession() { // Declares a function to get saved session data
  const raw = localStorage.getItem("sports-platform-session"); // Reads session from
local storage
  return raw ? JSON.parse(raw) : null; // Parses the stored JSON or returns null
} // Ends getStoredSession function

function storeSession(session) { // Declares a function to save session data
  localStorage.setItem("sports-platform-session", JSON.stringify(session)); // Stores
session as JSON text
} // Ends storeSession function

function clearSession() { // Declares a function to remove stored session data
  localStorage.removeItem("sports-platform-session"); // Deletes the stored session
} // Ends clearSession function
```

This block helps the application maintain navigation state and preserve user login details between page reloads.

Registration and OTP verification unit:

This unit handles user registration and phone OTP verification before account creation.

```
async function requestOtp() { // Declares an asynchronous function to request OTP
  const value = form.mobileNumber; // Reads the mobile number from the form
  if (!value) { // Checks if the mobile number is empty
    setOtpStatus("Enter mobile number first."); // Displays a validation message
    return; // Stops execution if input is missing
  } // Ends validation block
  setOtpLoading((current) => ({ ...current, phone: true })); // Enables loading indicator
for OTP request
  try { // Starts try block
    const result = await api("/auth/request-otp", { // Sends OTP request to backend
      method: "POST", // Uses POST method
      headers: { "Content-Type": "application/json" }, // Sets JSON header
      body: JSON.stringify({ channel: "phone", value }) // Sends phone number in JSON
format
    }); // Ends API call
    setOtpStatus(result.message); // Displays response message
    setPhoneOtpSent(true); // Marks OTP as sent
```

```
    setVerification((current) => ({ ...current, phoneVerified: false, phoneVerifiedToken:
"" })); // Resets previous verification
  } catch (error) { // Handles request errors
    setOtpStatus(error.message); // Displays error message
  } finally { // Executes after try or catch
    setOtpLoading((current) => ({ ...current, phone: false })); // Disables OTP loading
indicator
  } // Ends finally block
} // Ends requestOtp function

async function verifyOtp() { // Declares an asynchronous function to verify OTP
  const value = form.mobileNumber; // Reads mobile number
  const otp = form.phoneOtp; // Reads entered OTP
  if (!value || !otp) { // Checks if required inputs are missing
    setOtpStatus("Enter mobile number and phone OTP."); // Displays validation
message
    return; // Stops execution
  } // Ends validation block
  setOtpLoading((current) => ({ ...current, verifyPhone: true })); // Enables verification
loading state
  try { // Starts try block
    const result = await api("/auth/verify-otp", { // Sends OTP verification request
      method: "POST", // Uses POST method
      headers: { "Content-Type": "application/json" }, // Sets JSON header
      body: JSON.stringify({ channel: "phone", value, otp }) // Sends phone number and
OTP
    }); // Ends API call
    setOtpStatus(result.message); // Displays server response
    setVerification((current) => ({ // Updates verification state
      ...current, // Keeps previous values
      phoneVerified: true, // Marks phone as verified
      phoneVerifiedToken: result.verifiedToken // Saves verification token
    })); // Ends state update
  } catch (error) { // Handles verification errors
    setOtpStatus(error.message); // Displays error message
  } finally { // Executes after try or catch
    setOtpLoading((current) => ({ ...current, verifyPhone: false })); // Disables loading
state
  } // Ends finally block
} // Ends verifyOtp function
```

This block ensures that the phone number is verified before the player or coach account is registered.

Video analysis unit:

This unit performs browser-side analysis of the uploaded performance video. It calculates duration, resolution, brightness, motion, sharpness, and brightness consistency.

```
async function analyzeVideoFile(file) { // Declares a function to analyse the uploaded video
  const video = document.createElement("video"); // Creates a temporary video element
  video.preload = "metadata"; // Loads video metadata first
  video.muted = true; // Mutes the temporary video
  video.playsInline = true; // Enables inline playback on supported devices

  const objectUrl = URL.createObjectURL(file); // Creates a temporary URL for the selected file
  video.src = objectUrl; // Assigns the temporary URL to the video source

  try { // Starts analysis block
    await waitForEvent(video, "loadedmetadata"); // Waits for video metadata to load

    const durationSec = Number(video.duration || 0); // Gets video duration in seconds
    const width = Number(video.videoWidth || 0); // Gets video width
    const height = Number(video.videoHeight || 0); // Gets video height
    const sampleCount = durationSec > 12 ? 6 : 4; // Selects the number of sample frames
    const sampleTimes = Array.from({ length: sampleCount }, (_, index) => // Creates sample time list
      Math.max(0, Math.min(durationSec - 0.05, (durationSec * (index + 1)) / (sampleCount + 1))) // Keeps sample points valid
    ); // Ends sample time creation

    const canvas = document.createElement("canvas"); // Creates a canvas for frame capture
    const sampleWidth = 96; // Sets sample width
    const sampleHeight = 54; // Sets sample height
    canvas.width = sampleWidth; // Applies width to canvas
    canvas.height = sampleHeight; // Applies height to canvas
    const context = canvas.getContext("2d", { willReadFrequently: true }); // Gets drawing context
    if (!context) { // Checks if canvas context is available
      throw new Error("Canvas analysis unavailable."); // Throws error if canvas fails
    } // Ends validation

    const brightnessValues = []; // Creates an array for brightness values
    const motionValues = []; // Creates an array for motion values
    const sharpnessValues = []; // Creates an array for sharpness values
    let previousFrame = null; // Stores the previous frame

    for (const sampleTime of sampleTimes) { // Loops through sample times
      video.currentTime = sampleTime; // Moves video to selected time
      await waitForEvent(video, "seeked"); // Waits until seeking completes
```

```
    context.drawImage(video, 0, 0, sampleWidth, sampleHeight); // Draws video frame
to canvas
    const stats = sampleFrameStats(context, sampleWidth, sampleHeight,
previousFrame); // Computes frame statistics
    brightnessValues.push(stats.brightness); // Stores brightness value
    motionValues.push(stats.motion); // Stores motion value
    sharpnessValues.push(stats.sharpness); // Stores sharpness value
    previousFrame = stats.frame; // Updates previous frame
  } // Ends loop

  const average = (values) => values.reduce((sum, value) => sum + value, 0) /
(values.length || 1); // Declares average helper
  const standardDeviation = (values, mean) => // Declares standard deviation helper
    Math.sqrt(values.reduce((sum, value) => sum + (value - mean) ** 2, 0) /
(values.length || 1)); // Calculates standard deviation

  const averageBrightness = average(brightnessValues); // Calculates average
brightness
  const averageMotion = average(motionValues); // Calculates average motion
  const averageSharpness = average(sharpnessValues); // Calculates average sharpness
  const brightnessDeviation = standardDeviation(brightnessValues,
averageBrightness); // Calculates brightness consistency

  return { // Returns computed video metadata
    durationSec, // Returns duration
    width, // Returns width
    height, // Returns height
    fileSizeMb: file.size / (1024 * 1024), // Converts file size to megabytes
    brightnessScore: Math.max(0, Math.min(100, (averageBrightness / 255) * 100)), //
Normalises brightness score
    motionScore: Math.max(0, Math.min(100, averageMotion / 1.8)), // Normalises
motion score
    sharpnessScore: Math.max(0, Math.min(100, averageSharpness / 2.2)), //
Normalises sharpness score
    brightnessConsistency: Math.max(0, Math.min(100, 100 - brightnessDeviation *
1.4)) // Normalises brightness consistency
  }; // Ends returned object
 } finally { // Starts cleanup block
  URL.revokeObjectURL(objectUrl); // Releases temporary URL memory
 } // Ends cleanup block
} // Ends analyzeVideoFile function
```

This block simulates the extraction of meaningful video features before upload to the backend.

Backend authentication and storage unit:

This unit configures the server, reads and writes the local database, and manages token-based authentication.

```
import "dotenv/config"; // Loads environment variables
import cors from "cors"; // Imports CORS middleware
import express from "express"; // Imports Express framework
import fs from "fs"; // Imports file system module
import path from "path"; // Imports path module
import bcrypt from "bcryptjs"; // Imports bcrypt password hashing library
import jwt from "jsonwebtoken"; // Imports JSON Web Token library
import multer from "multer"; // Imports file upload middleware
import crypto from "crypto"; // Imports crypto library
import twilio from "twilio"; // Imports Twilio library
import { fileURLToPath } from "url"; // Imports helper for current file path

const __filename = fileURLToPath(import.meta.url); // Converts module URL to file
path
const _dirname = path.dirname(_filename); // Gets current directory
const rootDir = path.resolve(__dirname, ".."); // Gets root project directory
const dbPath = path.join(rootDir, "data", "db.json"); // Creates database file path
const uploadsDir = path.join(rootDir, "uploads"); // Creates upload directory path
const app = express(); // Creates Express application
const port = 4000; // Sets server port
const JWT_SECRET = process.env.JWT_SECRET || "sports-talent-secret"; // Loads
JWT secret or uses fallback
const ADMIN_PAGE_KEY = process.env.ADMIN_PAGE_KEY || "admin123"; //
Loads admin key or uses fallback
const otpStore = new Map(); // Creates in-memory OTP store

app.use(cors()); // Enables cross-origin requests
app.use(express.json({ limit: "10mb" })); // Enables JSON body parsing
app.use("/uploads", express.static(uploadsDir)); // Serves uploaded files publicly

function ensureDb() { // Declares a function to ensure database exists
  if (!fs.existsSync(dbPath)) { // Checks if database file exists
    fs.mkdirSync(path.dirname(dbPath), { recursive: true }); // Creates parent directory
if needed
    fs.writeFileSync(dbPath, JSON.stringify({ players: [], coaches: [], videos: [] }, null,
2)); // Creates empty database file
  } // Ends condition
} // Ends ensureDb function

function readDb() { // Declares database read function
  ensureDb(); // Ensures database exists first
  return JSON.parse(fs.readFileSync(dbPath, "utf-8")); // Reads and parses JSON
database
```

```
} // Ends readDb function

function writeDb(data) { // Declares database write function
  fs.writeFileSync(dbPath, JSON.stringify(data, null, 2)); // Writes formatted JSON data to file
} // Ends writeDb function

function signToken(user) { // Declares token creation function
  return jwt.sign({ id: user.id, role: user.role }, JWT_SECRET, { expiresIn: "7d" }); // Creates JWT token
} // Ends signToken function
```

This block prepares the backend environment and handles secure data storage and authentication support.

AI scoring and upload integration unit:

This unit converts video metadata into sports performance scores and saves the uploaded video details.

```
function buildAnalysis(videoMeta, sport, position, ageGroup) { // Declares analysis generation function
  const meta = videoMeta || {}; // Uses metadata or empty object
  const width = Number(meta.width || 0); // Gets width value
  const height = Number(meta.height || 0); // Gets height value
  const durationSec = Number(meta.durationSec || 0); // Gets duration value
  const fileSizeMb = Number(meta.fileSizeMb || 0); // Gets file size value
  const brightness = Number(meta.brightnessScore || 0); // Gets brightness score
  const motion = Number(meta.motionScore || 0); // Gets motion score
  const sharpness = Number(meta.sharpnessScore || 0); // Gets sharpness score
  const brightnessConsistency = Number(meta.brightnessConsistency || 0); // Gets consistency score

  const resolutionPixels = width * height; // Calculates total resolution pixels
  const resolutionScore = normalize(resolutionPixels, 640 * 360, 1920 * 1080) * 100; // Normalises resolution score
  const durationScore = normalize(durationSec, 8, 75) * 100; // Normalises duration score
  const bitrateMbps = durationSec > 0 ? (fileSizeMb * 8) / durationSec : 0; // Estimates bitrate
  const bitrateScore = normalize(bitrateMbps, 0.5, 8) * 100; // Normalises bitrate
  const balancedBrightness = 100 - Math.abs(brightness - 55) * 1.8; // Computes balanced brightness value
  const stabilitySignal = brightnessConsistency; // Assigns brightness consistency as stability

  const scores = { // Creates performance score object
```

```
    speedScore: metricScore(motion * 0.7 + durationScore * 0.15 + bitrateScore * 0.15),
// Calculates speed score
    accuracyScore: metricScore(sharpness * 0.45 + stabilitySignal * 0.35 +
balancedBrightness * 0.2), // Calculates accuracy score
    reactionTime: metricScore(motion * 0.6 + sharpness * 0.25 + bitrateScore * 0.15), //
Calculates reaction time
    movementEfficiency: metricScore(stabilitySignal * 0.4 + motion * 0.35 +
balancedBrightness * 0.25), // Calculates movement efficiency
    techniqueQuality: metricScore(sharpness * 0.45 + resolutionScore * 0.3 +
bitrateScore * 0.25), // Calculates technique quality
    staminaEstimation: metricScore(durationScore * 0.65 + motion * 0.2 +
stabilitySignal * 0.15) // Calculates stamina estimate
  }; // Ends score object

  const overallPerformanceScore = Math.round( // Calculates final overall performance
score
    Object.values(scores).reduce((sum, value) => sum + value, 0) /
Object.values(scores).length // Averages the metrics
  ); // Ends score calculation

  return { // Returns final analysis object
    ...scores, // Includes all calculated metric scores
    overallPerformanceScore, // Includes overall performance score
    analysisBasis: { // Includes supporting analysis values
      durationSec: Math.round(durationSec * 10) / 10, // Stores rounded duration
      resolution: width && height ? ${width}x${height} : "Unavailable", // Stores
resolution text
      fileSizeMb: Math.round(fileSizeMb * 100) / 100, // Stores rounded file size
      brightnessScore: metricScore(brightness), // Stores rounded brightness
      motionScore: metricScore(motion), // Stores rounded motion
      sharpnessScore: metricScore(sharpness), // Stores rounded sharpness
      brightnessConsistency: metricScore(brightnessConsistency), // Stores rounded
consistency
      ageGroup // Stores age group
    } // Ends basis object
  }; // Ends return object
} // Ends buildAnalysis function
```

This block acts as the simulated AI engine of the project by converting video properties into player performance metrics.

Video upload form integration:

This block integrates the player upload form with the browser-side analysis and backend storage.

```
async function handleSubmit(event) { // Declares form submission function
  event.preventDefault(); // Prevents page refresh on form submit
```

```
  if (!file || !file.type.startsWith("video/")) { // Checks if selected file is a valid video
    setStatus("Only video files are allowed."); // Displays validation error
    return; // Stops submission
  } // Ends validation

  setLoading(true); // Enables loading state
  setStatus("Analyzing Video..."); // Displays analysis progress message

  let analysisMeta; // Declares metadata variable
  let videoHash; // Declares video hash variable
  try { // Starts try block
    [analysisMeta, videoHash] = await Promise.all([analyzeVideoFile(file),
hashFile(file)]); // Runs analysis and hashing together
  } catch (error) { // Handles analysis error
    setLoading(false); // Disables loading state
    setStatus(error.message); // Displays error message
    return; // Stops process
  } // Ends catch block

  setStatus("Uploading Video..."); // Displays upload progress
  const formData = new FormData(); // Creates multipart form data
  formData.append("video", file); // Appends video file
  formData.append("sport", selectedSport); // Appends selected sport
  formData.append("ageGroup", ageGroup); // Appends selected age group
  formData.append("position", position); // Appends selected position
  formData.append("analysisMeta", JSON.stringify(analysisMeta)); // Appends
analysis metadata
  formData.append("videoHash", videoHash); // Appends video hash

  try { // Starts upload request
    const result = await api("/videos/upload", { method: "POST", body: formData },
token); // Sends data to backend
    setStatus(result.warning || "Video Successfully Submitted"); // Displays response
message
    onUploaded(result.video); // Sends uploaded video result to parent component
  } catch (error) { // Handles upload error
    setStatus(error.message); // Displays error message
  } finally { // Executes after success or failure
    setLoading(false); // Disables loading state
  } // Ends finally block
} // Ends handleSubmit function
```

This block connects the selected sport, age group, position, analysed metadata, and uploaded file to the backend processing unit.

Integration of the units

The software units are integrated in the following sequence:

- The frontend bootstrap unit loads the React application.
- The navigation and session unit manages page flow and preserves login sessions.
- The registration and OTP verification unit validates users before account creation.
- The player selects a sport and uploads a performance video.
- The browser-side video analysis unit extracts video metadata and generates pre-analysis values.
- The backend authentication and storage unit receives the request and stores the file and user data.
- The AI scoring unit converts the metadata into performance metrics.
- The player dashboard displays the uploaded video and generated scores.
- The coach dashboard filters player submissions and supports shortlist management.
- The admin view displays overall database snapshots.
- Therefore, the software integrates user management, security, media handling, simulated AI processing, and dashboard reporting into one complete sports talent assessment system.

6.4 Simulation

Simulation allows the developer to test the system behavior without needing a real AI model or additional hardware. In this project, the simulation is not a circuit simulation or microcontroller simulation. Instead, it is a software simulation of AI-based sports performance assessment.

The project simulates AI analysis using deterministic rules derived from uploaded video metadata such as:

- duration
- resolution
- brightness
- motion
- sharpness
- brightness consistency

These values are processed by the backend to generate:

- speed score
- accuracy score
- reaction time
- movement efficiency
- technique quality
- stamina estimation
- overall performance score
- Simulation procedure used in this project
- The player registers and logs in to the web application.
- The player uploads a sports performance video.

- The browser analyses the video frames and extracts metadata.
- The backend simulates AI scoring using predefined formulas.
- The final output is displayed in charts, score bars, and review sections.
- The coach can review player performance and shortlist players.
- The admin can inspect stored records through the admin dashboard.

Why this simulation is appropriate:
This simulation method is suitable because the project is a web-based software platform, not an electronic hardware project. Therefore, tools like LTSpice, Proteus, Wokwi, or MATLAB/Simulink are not the primary simulation tools for this system. Instead, the system uses internal software logic to simulate the output of an AI sports assessment engine.

This approach helps to:
- test the complete workflow before implementing real AI models
- validate frontend and backend integration
- demonstrate scoring and dashboard functionality
- reduce development cost and complexity
- support MVP development and product validation
- Running the simulation

The local simulation of the complete software system can be performed using:
- npm run install:all
- npm run dev:server
- npm run dev:client

After running the commands:
- Open http://localhost:5173
- Register a player or coach
- Verify OTP if Twilio is configured
- Upload a sports performance video
- Observe the generated simulated AI analysis
- Review player data through coach and admin views

References:
- Project README, /Users/hemanth/Desktop/my pro/README.md
- Frontend entry file, /Users/hemanth/Desktop/my pro/client/src/main.jsx
- Frontend main application logic, /Users/hemanth/Desktop/my pro/client/src/App.jsx
- Backend server logic, /Users/hemanth/Desktop/my pro/server/src/index.js
- Frontend package configuration, /Users/hemanth/Desktop/my pro/client/package.json
- Backend package configuration, /Users/hemanth/Desktop/my pro/server/package.json
- If you want, I can now make this much more human and academic sounding, so it reads like a proper report submission instead of a technical draft.

Chapter 7

Evaluation and Results

7.1 Test points

The following test points were considered for evaluating the AI Sports Talent Assessment Platform:

- User registration for player and coach accounts
- Phone OTP request and verification process
- User login with valid and invalid credentials
- Session storage and user authentication
- Sports selection and navigation between pages
- Performance video upload with valid video format
- Rejection of invalid file types during upload
- Client-side video analysis for metadata extraction
- Backend AI score generation from uploaded video metadata
- Display of analysis results in charts and score bars
- Player dashboard retrieval and display of uploaded records
- Coach dashboard filtering and player shortlisting
- Admin database access and summary display
- Duplicate video detection using video hash
- Video deletion from player dashboard

7.2 Test plan

The test plan was prepared to verify whether each core module of the system performs correctly under normal and invalid conditions.

Test ID	Test Description	Input/Condition	Expected Result
TP1	Register player account	Valid player details	Player account created successfully
TP2	Register coach account	Valid coach details	Coach account created successfully
TP3	OTP request	Valid mobile number	OTP sent successfully
TP4	OTP verification	Correct OTP	Phone verified successfully

TP5	Login test	Valid email and password	Login successful
TP6	Invalid login test	Incorrect password	Error message displayed
TP7	Video upload	Valid video file	Valid video file
TP8	Invalid upload	Non-video file	Upload rejected
TP9	Video analysis	Valid uploaded video	Metadata extracted and processed
TP10	AI scoring	Valid analysis metadata	Scores generated correctly
TP11	Player dashboard	Logged-in player	Uploaded videos displayed
TP12	Coach filtering	Sport, position, score filters	Matching players displayed
TP13	Shortlist feature	Add/remove player	Dashboard updated correctly
TP14	Admin database test	Valid admin key	Summary and records displayed
TP15	Duplicate upload test	Previously uploaded video	Duplicate warning displayed

7.3 Test Result

The test results show that the system performed successfully for the major functional requirements.

Test ID	Actual Result	Status
TP1	Player registration completed successfully	Pass
TP2	Coach registration completed successfully	Pass
TP3	OTP request worked when configuration was available	Pass
TP4	OTP verification returned verification token	Pass
TP5	Valid login redirected user correctly	Pass
TP6	Incorrect password produced error message	Pass
TP7	Valid video uploaded and stored successfully	Pass
TP8	Invalid file type was rejected	Pass
TP9	Video duration, resolution, brightness, motion, and sharpness were extracted	Pass
TP10	AI metric scores and overall score were generated	Pass
TP11	Player dashboard displayed uploaded video history	Pass
TP12	Coach filters returned relevant player records	Pass
TP13	Selected players were added and removed successfully	Pass
TP14	Admin page showed players, coaches, and videos summary	Pass
TP15	Duplicate upload displayed prior-analysis warning	Pass

The obtained results confirm that the developed platform is functionally stable for MVP-level deployment and demonstration.\

7.4 Insights

The following insights were obtained from the evaluation:

- The system successfully integrates frontend, backend, authentication, upload handling, and simulated AI analysis into a single workflow.
- Client-side video analysis reduces backend processing by extracting useful metadata before upload.
- Deterministic scoring provides a practical simulation approach for testing the platform before integrating a real AI model.
- OTP verification improves account validation and increases reliability of registration data.
- The player dashboard gives clear feedback through scores, analysis basis, and performance review.
- The coach dashboard is effective for filtering and shortlisting players according to sport, position, and score.
- Duplicate video detection helps avoid redundant analysis and improves efficiency.
- The current system is suitable for demonstration, academic presentation, and MVP validation.
- A future improvement would be replacing simulated scoring with real machine learning or computer vision models.
- Another improvement would be moving from local JSON storage to a full database such as MongoDB or PostgreSQL for better scalability.

Chapter 8

Social, Legal, Ethical, Sustainability, and Safety Aspects

8.1 Social Aspects

Social aspects examine how the platform influences communities and human interactions.
Positive impacts:

- Democratizes access to sports talent assessment, enabling athletes from rural and underserved regions to participate using smartphones.
- Promotes inclusivity and fairness by reducing bias in scouting.
- Encourages grassroots participation, strengthening community engagement and youth involvement in sports.

Negative impacts:

- Risk of digital divide if athletes lack smartphones or stable internet.
- Potential misuse of data for non-sporting purposes, leading to privacy concerns.

Case Study (AI impact): AI adoption in sports has improved performance analysis but also raised concerns about algorithmic bias and over-reliance on technology. Similar risks apply here if datasets are not diverse.

8.2 Legal Aspects

Legal aspects ensure compliance with data protection and intellectual property laws.

- **Data Privacy**: Athlete data must comply with frameworks such as GDPR (EU) and India's Digital Personal Data Protection Act (DPDPA, 2023).
- **Consent & Ownership**: Athletes must provide informed consent; ownership of performance data should remain with them.
- **Compliance Challenges**: Cloud deployment requires adherence to cybersecurity standards and cross-border data regulations.

8.3 Ethical Aspects

Ethics guide fairness, accountability, and respect for human dignity.

- **Fairness**: AI models must be trained on diverse datasets to avoid bias.
- **Transparency**: Athletes and coaches should understand how metrics are calculated.
- **Accountability**: Developers and institutions are responsible for errors in AI predictions.
- **Quality of Life**: The project enhances opportunities for athletes but must avoid depersonalization or addictive use.
- **Ethical Analysis of Illegal Activities**: If the platform is misused (e.g., falsifying data), ethical analysis shows such actions harm fairness, violate trust, and undermine legality.

8.4 Sustainability Aspects

Sustainability focuses on environmental and social responsibility.

- **Efficient Resource Use**: Smartphones reduce the need for costly infrastructure, minimizing material waste.
- **Durability**: Cloud scalability ensures long-term usability without excessive hardware.
- **Environmental Impact**: Digital assessments reduce travel and physical infrastructure, lowering carbon footprint.
- **Social Sustainability**: Promotes equal opportunities and fair labor practices by extending access to marginalized athletes.

8.5 Safety Aspects

Safety ensures secure, reliable, and harm-free operation.

- **Data Security**: Encryption, authentication, and role-based access protect athlete data.
- **System Reliability**: Cloud redundancy and load balancing prevent failures during large-scale use.
- **User Safety**: Athletes only use smartphones, avoiding invasive or risky hardware.
- **Cybersecurity**: Continuous monitoring prevents breaches and unauthorized access.

Chapter 9

Conclusion

This project has managed to illustrate how artificial intelligence, computer vision and mobile computing can be incorporated into a single platform to democratize the process of talent evaluation in sports.The proposed system can empower athletes, coaches, and scouts alike by overcoming the shortcomings of the traditional scouting systems: subjectivity, geographic bias, and reliance on resources, to offer a clear, scalable and inclusive solution that empowers each player, coach and scout.

The platform provides real-time, bias-free analysis of major performance indicators (stride length, jump height reaction time, and posture) through data capture over smartphones, cloud-based artificial intelligence processing, and interactive dashboards.These are measures that are compared to the global standards, hence objectivity and impartiality in talent recognition.Individualized feedback and tracking of progress are beneficial to the athlete, and structured analytics are accessible to coaches and scouts assisting them in making data-intensive recruitment choices.

The architecture of the system, including the modular backend services, secure APIs, and scalable cloud infrastructure, is reliable, flexible, and inclusive to various sporting scenarios.Notably, the project is in line with some of the United Nations Sustainable Development Goals (SDGs), such as good health and well-being, enhancing quality education through skills training, gender equality by eradicating biasness, promoting innovation in sports analytics, and reducing inequalities through the spread of opportunities to the underserved.

Although the project has strengths, it recognizes some weaknesses, including reliance on the quality of smartphone hardware, the lack of connectivity in remote locations, and the necessity to constantly retrain AI models to respond to different sports and demographics.These issues mark the future work opportunities such as offline functionality, increased sports coverage, and inclusion of psychological and cognitive assessment modules.

Altogether, the AI-Powered Mobile Platform is a revolutionary move towards building a globally interconnected sports talent discovery.The project fills the gap in the participation between grassroots and elite scouting, creates inclusivity, and leads to the development of sports analytics as a field, by enabling accessibility, objectivity, and scalability.It prepares a solid ground towards the future innovations which can further promote fairness, transparency and opportunity in the field of sport.

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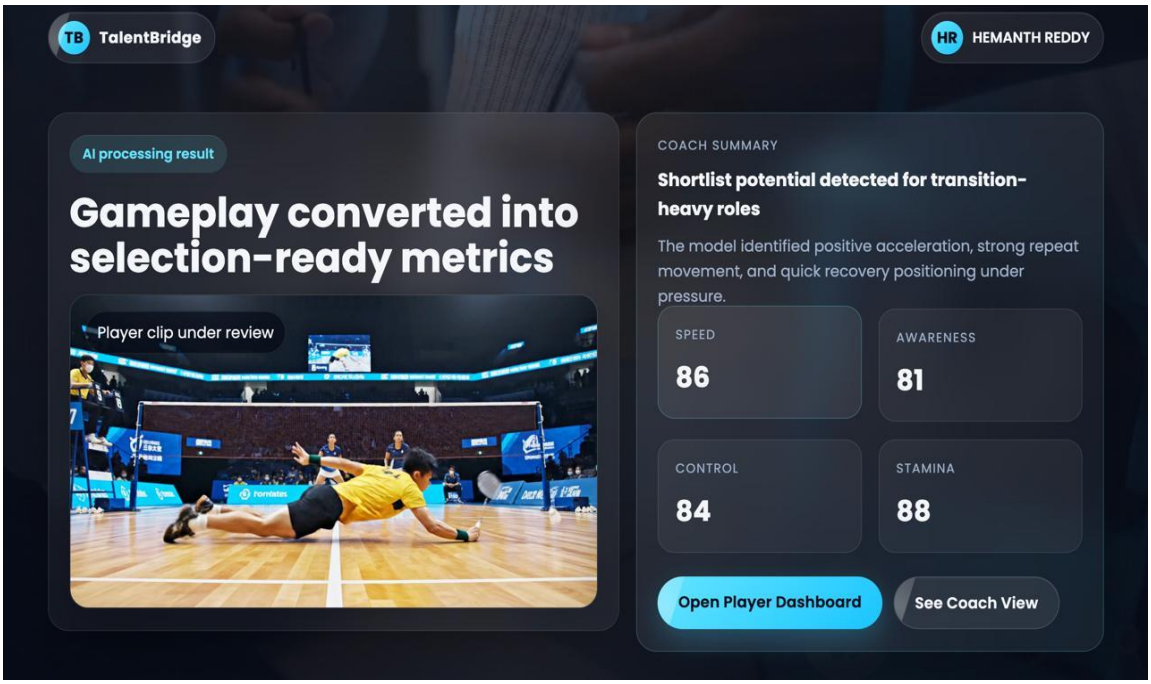
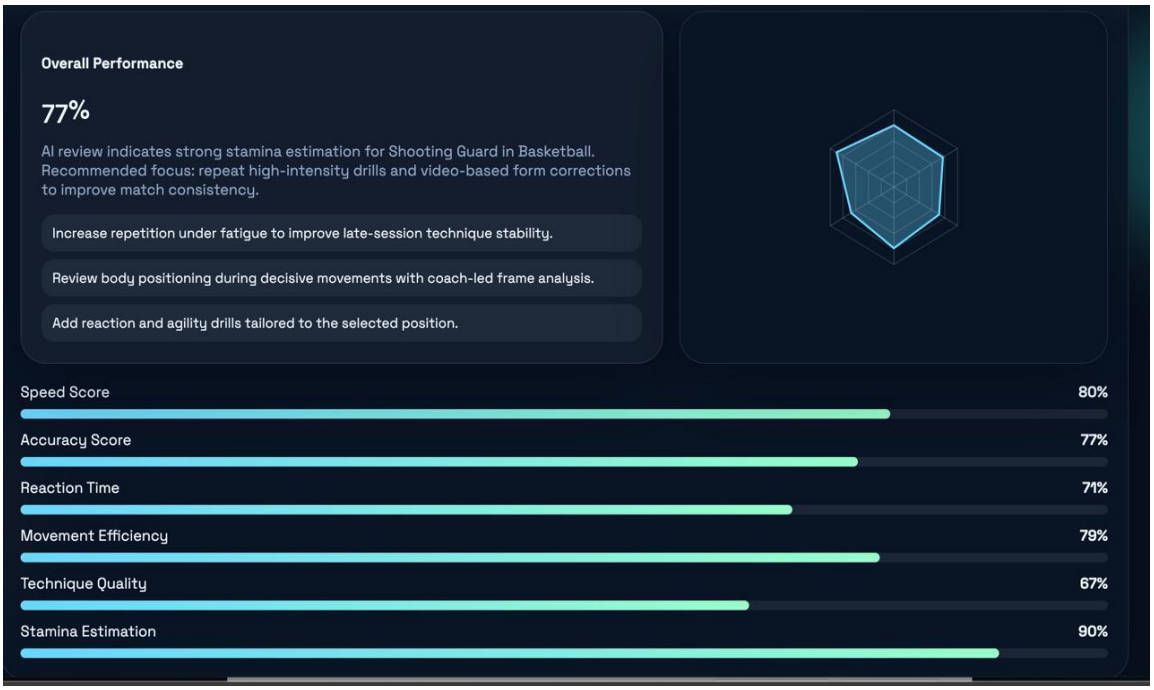
Appendix


TB TalentBridge
HR HEMANTH REDDY
AI processing result
Gameplay converted into selection-ready metrics
Player clip under review
COACH SUMMARY
Shortlist potential detected for transition-heavy roles
The model identified positive acceleration, strong repeat movement, and quick recovery positioning under pressure.
SPEED
86
AWARENESS
81
CONTROL
84
STAMINA
88
Open Player Dashboard
See Coach View


Overall Performance
77%
AI review indicates strong stamina estimation for Shooting Guard in Basketball. Recommended focus: repeat high-intensity drills and video-based form corrections to improve match consistency.
Increase repetition under fatigue to improve late-session technique stability.
Review body positioning during decisive movements with coach-led frame analysis.
Add reaction and agility drills tailored to the selected position.
Speed Score 80%
Accuracy Score 77%
Reaction Time 71%
Movement Efficiency 79%
Technique Quality 67%
Stamina Estimation 90%